Class 7
PHYSICS

Table of Content

1.	Units and Dimensions	1 – 12
2.	Vectors	13 – 15
3.	Kinematics	16 – 52
4.	Types of forces	53 – 55
5.	Newtons Laws of Motion	56 – 76

1. UNITS & DIMENSIONS

WORKSHEET - 1

Level – I

1)A	2) C	3) B	4) C	5) A	6) B
7) C	8) C				

Level – II

1) D	2) D	3) C	4) D	5) B	6) B
7) B	8) D	9) D			

Level – III

10) A,D	11) A	12) A,B,C	13) C	14) D	15) C
16) A,B	17) A	18) C	19) D	20) B	21) A
22) D	23) D	24) D	25) a-r; b-t; c-s; d-p,q		
26) a-q; b-r; c-p; d-s		27) C,D	28) B		29) A

WORKSHEET – 2

Level – I

1) B	2) A	3) A	4) B	5) A	6) B	7) C

Level – II

1) B	2) D	3) B	4) A	5) B	6) A
7) D	8) B	9) B	10) B	11) A	12) B
13) B	14) A				

12. $N_1U_1 = N_2U_2$

$AB = CD$

$\frac{N_1}{N_2} = \frac{U_2}{U_1}, \frac{A}{C} = \frac{D}{B}$

$100 = D/B$

$\frac{1}{100} = \frac{B}{D}$

Level – III

15) A,B,C	16) A,B,C,D	17) A,B	18) C	19) B
20) C	21) B	22) B	23) A,B	

WORKSHEET - 3

Level – I

1) C	2) B	3) A	4) D	5) D	6) C
7) B	8) B	9) B			

Level – II

1) B	2) A	3) B	4) B	5) 3	6) 1
7) D	8) B	9) C			

Level – III

10) D	11) A	12) D	13) D	14) D	15) A
16) C	17) C	18) D	19) D	20) C	
21) a-s; b-q,r; c-p,q,r,s; d-p			22) a-q; b-t; c-s; d-r		23) 1,2,4
24) B	25) A	26) B,D	27) A,B,C	28) A,B,D	

WORKSHEET - 4

Level – I

1. B	4. D	7. C	10.B
2. A	5. B	8. B	
3. C	6. C	9. D	

9. $1\text{ cm} = \text{mm}$

$1 \times 10^{-2}\text{ m} = x \times 10^{-3}\text{ m}$

$\frac{10^{-2}}{10^{-3}} = x$

$x = 10$

Level – II

1) C	2) A	3) A	4) C	5) A	6) A
7) A	8) A	9) C	10) C	11) A	12) D
13) C	14) A	15) C	16) B	17) A	18) C
19) B	20) C	21) D	22) D	23) A	24) C

13. $1\text{ kg} = \mu g$

$10^3\text{ g} = X10^{-6}\text{ g}$

$10^3 = X10^{-6}$

$10^3 \times 10^6 = X$

$X = 10^9$

14. $1\mu s = ms$

$10^{-6}s = X10^{-3}s$

$10^{-6} = X10^{-3}$

$10^{-6}10^3 = X$

$10^{-3} = X$

15. $1\text{hm} = ---\text{Gm}$

$1 \times 10^2\text{ m} = x \times 10^9\text{ m}$

$\frac{10^2}{10^9} = x$

$x = 10^{-7}$

16. $\frac{1mm}{1\mu s} = ---\frac{mm}{ms}$

$\frac{1 \times 10^{-3}\ \text{m}}{1 \times 10^{-6}\ \text{s}} = x \times \frac{10^{-3}}{10^{-3}\ \text{s}}$

$10^{-3} = x$

26. 1 mm $= --$ km

$1 \times 10^{-3} = x \times 10^3$ m

$\frac{10^{-3}}{10^3} = x$

$x = 10^{-6}$

1 mm $= ---$nm

1×10^{-3} m $= x \times 10^{-9}$ m

$\frac{10^{-3}}{10^{-9}} = x$

$x = 10^6$

1 mm $= --$ m

1×10^{-3} m $= x \times 10^{-12}$ m

$\frac{10^{-3}}{10^{-12}} = x$

$x = 10^9$

33. (a) 1 mm $= --$ m

1×10^{-3} m $= x$ m

$x = 10^{-3}$

b) 1ng $= --$ mg

1×10^{-9} g $= x \times 10^{-3}$ g

$x = 10^{-6}$

c) 1μg $= --$ gram

$1 \times 10^{-6} g = xg$

$x = 10^{-6}$

d) 1 ms $= --$ s

$1 \times 10^{-3} s = xs$

$x = 10^{-3}$

Level – III

25) A,C,D 26) B,D 27) A,B,C 28) A 29) a-q,s; b-t; c-p; d-r

30) A 31) B 32) B 33) a-q; b-p; c-p; d-q

34) a-q; b-s; c-r; d-p

WORKSHEET - 5

Level – I

1) C	2) C	3) A	4) C	5) D	6) C
7) C	8) A	9) D	10) C	11) C	

Level – II

1) D	2) B	3) D	4) C	5) D	6) B
7) D	8) B	9) B	10) B	11) A	12) B
13) C	14) B	15) C	16) B	17) A	18) A
19) A	20) D	21) D	22) A	23) D	

4. $p = \frac{w}{t} = \frac{f \times s}{t} = \frac{m \times a \times s}{t} = \frac{kg \times ms^{-2} \times m}{s}$

$= \frac{[M \times LT^{-2} \times L]}{[T]}$

$P = [ML^2T^{-2-1}]$

$P = [M^1L^2T^{-3}]$

Dimensions of time in power is T^{-3}

6. $p = \frac{w}{t}, L^a$ is = ?

Power of Length is what

WKT $p = \frac{F \times S}{t} = \frac{m \times a \times s}{t}$

$= \frac{kg \times ms^{-2} \times m}{t}$

$= \frac{[ML^2T^{-2}]}{[T]}$

$[ML^2T^{-2-1}]$

$p = [M^1L^2T^{-3}]$

Power of $L = L^2 = 2$

15. $(p + a/v^2)(v - b) = RT$

⇒ P is pressure v is volume, and T is temperature.

$[P] = [F/A] = [MLT^{-2}/L^2] = [MLT^{-1}/L^2]; [V] = [L^3]$

We cannot add or subtract quantities of different dimensions.

Therefore $P = [a/V^2] \Rightarrow [a] = [PV^2] = [ML^{-1}\ T^{-2}\ L^6] = [ML^5\ T^{-2}]$

16. $a = [M^0T^2]$

17. $b = [M^{-1}L^0T^4]$

18. $\frac{a}{b} = \frac{[M^0T^2]}{[M^{-1}L^0T^4]} = [M^1L^0T^{-2}]$

$= [M^{0+1}L^0T^{2-4}]$

$\frac{a}{b} = [M^1L^0T^{-2}]$

23. $E = 100\text{JE}$

$M^1 = 2m$

$T^1 = 2t$

$L^1 = \frac{l}{2}$

Energy formula (Kinetic or potential energy)

$E = \frac{1}{2} \times m \times v^2$

$= \frac{1}{2} \times 2m \times \left(\frac{s}{t}\right)^2$

$100E = \frac{1}{2} \times 2m \times \left(\frac{\frac{l}{2}}{2t}\right)^2$

$= \frac{1}{2} \times 2m \times \frac{\frac{l^2}{4}}{4t^2}$

$100E = \frac{1}{2} \times 2 \times \frac{1}{16} \times m\frac{l^2}{t^2}$

$8 \times 100E = \frac{1}{2}m\left(\frac{l}{t}\right)^2$

$800E = \frac{1}{2}mv^2$

Comprehension Type:

$\mu = A + \frac{B}{\lambda} + \frac{C}{\lambda^2}$

According to principle of Homogenity of dimensione

$\mu = A, \mu = \frac{B}{\lambda}, \mu = \frac{C}{\lambda^2}$

λ = Wavelength = meter = $[L]$

μ = constant.

Level – III

24) 1,2 25) 1 26) 2 27) 3 28) a-q,r; b-p,s; c-r; d-q,r

29) 1,4 30) 1,2,3 31) 4

25. Dimensional formula for ' A ' is

$\mu = A$

μ = Constan t so $[M^0L^0T^0]$

26. Dimensional formula of ' B ' is

$\mu = \frac{B}{\lambda} \Rightarrow B = \lambda\mu$

$B = \text{met}$

$B = [L]$

$B = [M^0L^1T^0]$

27. Dimensional formula of ' C ' is

$\frac{\mu}{1} = \frac{c}{\lambda^2}$

$c = \mu\lambda^2$

$c = [L^2]$

$c = [M^0L^2T^0]$

28. $a) f = \left(\frac{a}{t}\right) + bt^2$

dimensions of a is

$f = \frac{a}{t}$

$a = f \times t = m \times a \times t = kg \times ms^{-2} \times s$

$= [MLT^{-2} \times T]$

$= [MLT^{-2+1}]$

$a = [MLT^{-1}]$

b) $v = at + \frac{b}{t+c}$

$v = at, v = \frac{b}{t+c}$ or $v = \frac{b}{t}$

$a = \frac{v}{t}$

$a = \frac{ms^{-1}}{s} = \frac{[LT^{-1}]}{[T^1]} = [LT^{-1-1}]$

$a = [LT^{-2}]$

$b = v \times t$

$= ms^{-1} \times s$

$[LT^{-1} \cdot T^{-1}] = [LT^{-1+1}] = [L]$

c) $f = at + bt^2$

$f = bt^2,\ f = at$

$b = \frac{f}{t^2 = \frac{m \times a}{t^2}} = \frac{Kgms^{-2}}{s^2} = \left[\frac{MLT^{-2}}{T^2}\right] = [MLT^{-2-2}]$

$b = [MLT^{-4}]$

d) $f = at^{-1} + bt^2$

$f = \frac{a}{t} + bt^2$

$a = f \times t = m \times at = kgms^{-2} \times s = [MLT^{-2+1}] = [MLT^{-1}]$

$b = \frac{f}{t^2} = \frac{[MLT^{-2}]}{[T^2]} = [MLT^{-2-2}] = [MLT^{-4}]$

WORKSHEET - 6

Level – I

1) 3	2) 4	3) 3	4) 1	5) 4	6) 1
7) 4	8) 3	9) 2	10) 1	11) 3	

Level – II

1) A	2) B	3) C	4) A	5) B	6) C
7) B	8) B	9) C	10) D	11) C	12) C
13) A	14) C	15) B	16) D	17) A	18) C
19) A	20) A	21) B			

1. $1\text{ m} = 100\text{ cm}$

   $500\text{ cm} = 5\text{ m}$

2. $3\text{ cm} = x\text{ nm}$

   $3 \times 10^{-2}\text{ m} = x10^{-9}\text{ m}$

   $3 \times 10^{-2} \times 10^{9} = x$

   $3 \times 10^{7} = x$

   $0.3 \times 10^{8} = x$

3. $500\text{ kg} = x\text{mg}$

   $500 \times 10^{3}\text{ g} = x10^{-3}\text{ g}$

   $5 \times 10^{5} \times 10^{3} = x$

   $x = 5 \times 10^{8}$

4. 60 minutes = sec

   1 minute = 60 s

   60 minute = $60 \times 60 = 3600$ s

5. $40\text{ cm}^3 = 40 \times (10^{-2}\text{ m})^3 = 40 \times 10^{-6}\text{ m}^3$

6. $7\text{ m}^3 = 7 \times 10^{9}\text{ mm}^3$

7. $36\text{kmph} = 36 \times \frac{5}{18} = \frac{36\times5}{18} \times 100\frac{\text{cm}}{\text{s}} = 1000\frac{\text{cm}}{\text{s}}$

8. $5\mu N = 5 \times 10^{-4}$ kilo dyne

9. 5 Kilo newton = dyne

   $1\text{ N} = 10^{5}$ dyne

   5 Kilo newton $= 5 \times 10^{3}$ N

   5 Kilo newton $= 5 \times 10^{3} \times 10^{5}$ dyne

   $= 5 \times 10^{8}$ dyne

   5 Kilo newton $= 5 \times 10^{8}$ dyne

11. 37MJ = Herg

    $37MJ = 37 \times 10^{6}J$

    $= 37 \times 10^{6} \times 10^{7} \times 10^{6}$Herg

$37MJ = 37 \times 10^{19}$Herg

12. 1 kg = $x\mu g$

1×10^3 g = $x10^{-6}$ g

$x = 10^9$

16. 1 mean solarday = 1×24hr

$= 1 \times 24 \times 60$ minute (or) 1440 min

$= 1 \times 24 \times 60 \times 60$ s(or) 86400 s

1 mean solarday = 86400 s

19. 150kmph = x m/s

$150 \times \frac{5}{18} = x$

$x = 41.66$

20. 20×10^{-9} g = $x10^3 \times 10^3$ g

$20 \times 10^{-15} = x$

$x = 2 \times 10^{-14}$

21. 75kmph = $x$mph

75/1.609mph = $x$mph

75/1.609 = x

$x = 46.6$

Level – III

22) a-r; b-s; c-p; d-t;	23) A,B,C,D	24) C	25) C	26) C
27) a-p; b-q,r; c-s; d-t	28) B	29) A	30) C	

WORKSHEET - 7

Level – I

1) D	2) A	3) C	4) C	5) D	6) A
7) A	8) B	9) B	10) C	11) A	

1. 1.Two consecutive threads of screw, along axis of the screw
   2.The smallest division. on the main scale
2. Sleeve cylinder
3. Ratchet
4. Least count
5. screw gauge
6. Sleeve cylinder
7. P
8. Is on the base line of measurement
9. negative zero error.
10. Positive zero error

11. Micrometre.

Level – II

1) D	2) B	3) D	4) B	5) A	6) B
7) D	8) C	9) C	10) C	11) C	12) D
13) D	14) D	15) B	16) D	17) D	18) A
19) B					

1. 0.01 cm
2. 0.1 mm (or) 0.01 cm (or) 0.0001 m
3. 1 mm (or) 0.1 cm
4. 0.001 cm (or) 0.01 mm
5. $\text{Pitch} = \frac{\text{dis tan cetravelled by the screw}}{\text{Noof completerotations}}$

   $= \frac{5 \text{ mm}}{5} = 1 \text{ mm}$

   Pitch = 1 mm.
6. $\text{L.C} = \frac{\text{Pitchof the screw}}{\text{Noof head scale divisions}}$

   $= \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$

   L.C = 0.001 cm
7. $\text{Pitch} = \frac{\text{dis tan cetravelled by the screw}}{\text{Noof completerotations}}$

   $= \frac{2 \text{ mm}}{4} = \frac{1}{2} \text{ mm}$

   pitch = 0.05 cm.
8. $\text{L.C} = \frac{\text{Pitch}}{\text{Noof head scaledivisions}}$

   $= \frac{0.05 \times 100}{50 \times 100} = \frac{5}{5000} = \frac{1}{1000}$

   $= 1 \times 10^{-3} = 0.0001$ cm

   L.C = 0.001 cm
9. $L.C = \frac{\text{Pitch}}{\text{No of head scaledivisions}} = \frac{1 \text{ mm}}{100} = 0.001 \text{ cm}$
10. $\text{Pitch} = \frac{\text{Dis tan cetravelled by the screw}}{\text{Noof completerotation made.}}$

    $\text{Pitch} = \frac{Y}{Z}$

    $\text{Least count} = \frac{\text{Pitch}}{\text{Noof head scaledivisions}}$

    $= \frac{Y}{z}$

    $L.C = \frac{Y}{Zn}$

11. $L.C = \frac{\text{Pitch}}{\text{Noof head scaledivisions}}$

$= \frac{1 \text{ mm}}{100} = 0.01 \text{ cm}$

$L.C = 0.001 \text{ cm}$

12. Diameter of the screw gauge

$= M.S.R + (C.S.R + L.C)$

$= 0.5 + (35 \times 0.001)$

$d = 0.535 \text{ cm}$

13. Reading $= M.S.R + (C.S.R \times L.C)$

$= 2 + (62 \times 0.01)$

$= 2 + 0.62$

Re ading = 2.62 mm

14. $L.C = \frac{\text{Pitch}}{\text{Noofdivisions}} = \frac{1 \text{ mm}}{200} = 0.005 \text{ mm}$

$L.C = 0.0005$ cm.

15. Diameter $= M.S.R + (C.S.R \times L.C)$

$= 0.5 + (35 \times 0.0005)$

$= 0.5 + 0.0175$

$d = 0.5175$

$d = 0.5175$

$r = \frac{0.5175}{2}$

$r = 0.25875 \text{ cm}$

16. Pitch = 1 mm

$$L.C = \frac{\text{Pitch}}{\text{Noof head scale divisions}}$$

$L.C = 0.001 \text{ cm}$

17. Diameter = M.S.R $+(C.S.R \times L.C)$

$= 0.2 + (45 \times 0.001)$

$= 0.2 + 0.045$

$d = 0.245$ cm.

18. $d = M.S.R + (C.S.R + L.C)$

$= 1 \text{ mm} + (27 \times 0.01 \text{ mm})$

$= 1 + 0.27$

$= 1.27 \text{ mm}$

$d = 0.127 \text{ cm}$

19. Correct diameter reading = diameter - error

$= 0.127 - 0.005$

$= 0.122$ cm.

Level – III

20) C,D 21) A 22) A 23) B 24) C

25) a-s; b-q; c-r; d-p 26) 100. 27) A 28) A 29) A,B,C,D

30) A,B

20. pitch $= \frac{4mm}{4} = 1$ mm $= 0.1$ cm

$L.C = \frac{1 \text{ mm}}{50} = 0.02$ mm

22. Pitch $= \frac{\text{Dis tan cetravelled by screw}}{\text{No of complete rotations}}$

$= \frac{1 \text{ mm}}{2} = 0.5$ mm

23. The least count of the screw gauge is

$L.C = \frac{\text{Pitch}}{\text{Noof divisions onthehead scale}}$

$L.C = \frac{0.5 \text{ mm}}{50} = 0.01$ cm

$L.C = 0.001$ cm

24. $M.S.R = 0.5$ mm $= 0.05$ cm

$L.C = 0.001$ cm

C.S.R $= 27$

diameter $= MSR + (C.S.R \times LC)$

$= 0.05 + (27 \times 0.001)$

$= 0.05 + 0.027$

$= 0.077$ cm.

25. (a) Pitch $= \frac{1}{20}$ cm

$= 0.05$ cm

(b) $M.S.R = M.S.D \times$ pitch

$= 3 \times 0.05$

$= 0.15$ cm

$d = M.S.R + (C.S.R \times L.C)$

$= 0.15 + (24 \times 0.001)$

$= 0.15 + 0.024$

$= 0.174$ cm

(d) $d = M.S.R + (C.S.R \times L.C) +$ error

$= 0.174 + 0.008 = 0.182$ cm

26. Pitch $= 1$ mm

$L.C = 0.001$ cm

no. of divisions $= \frac{Pitch}{L.c} = \frac{1}{0.001 \text{ cm}} = \frac{1}{0.01 \text{ mm}} = 100$ mm

$= 100$

29. Error $= 7$ divisions

$M.S.R = 2 \times 0.1 = 0.2$ cm

$C.S.R = 79\ M.S.D = 10$

$C.S.P = 100$

(1) Pitch $= \frac{1}{10} = 0.1$ cm

Leastcount $= \frac{\text{Pitch}}{\text{Noof head scaledivisions}}$

$= \frac{0.1}{100} = 0.001$ cm

(2) Leastcount $= \frac{\text{pitch}}{\text{Noof head scale}}$

$= \frac{0.05}{50}$

$= 0.001$ cm

(3) diameter $= M.S.R + (C.S.R \times L.C)$

$= 0.2 + (79 \times 0.001)$

$= 0.2 + 0.079$

$= 0.279$

4. diameter + error

$= 0.279 + 0.007$

$= 0.286$ cm

2. VECTORS

WORKSHEET-1

Level – I

1) A	2) D	3) B	4) D	5) C	6) D	7) C
8) C	9) C	10) C				

Level – II

1) C	2) D	3) B	4) B	5) D	6) B	7) B
8) A	9) C	10) C	11) C	12) D	13) A	14) B
15) D	16) D	17) B	18) A	19) C	20) C	21) C

2. A vector is lying in XY plane it contains unit vectors $ and $ and magnitude of the vectors can be represented by (1) $\sqrt{x^2+y^2}$ (2) $10\sqrt{2} = \sqrt{10^2 \times 10^2}$ comparing (1) and (2) component of given vector along x-axis is 10 units and component of given vector along y-axis is 10 units so vector can be represented by 10$ + 10$

3. Magnitude of a vector $a\hat{\imath} + b\hat{\jmath} + c\hat{k}$ is $\sqrt{a^2+b^2+c^2}$
Magnitude of a vector $5\hat{\imath} + \hat{\jmath} - 2\hat{k}$ is $\sqrt{5^2+1^2+2^2}$
$= \sqrt{25+1+4} = \sqrt{30}$

4. Magnitude of unit vector is 4 unit
$1 = \sqrt{(0.5)^2 + (0.8)^2 + c^2}$ [$\therefore$ squaring on both sides]
$1 = 0.25 + 0.64 + c^2$
$c^2 = 0.11$
$c = \sqrt{0.11}$

7. Position vector of a particle in a rectangular co-ordinate system is (x, y, z) then its position vector is $x\hat{\imath} + y\hat{\jmath} + z\hat{k}$. Given position of a particle in rectangular coordinate system is (3,2,5) then its position vector is $3\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$.

8. Magnitude of given vector is $\sqrt{x^2+y^2}$

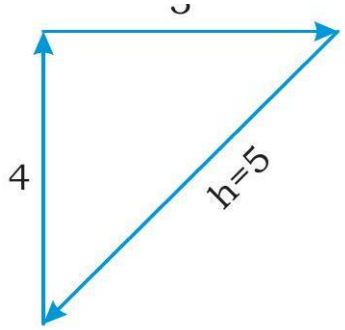
$$= \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2}+\frac{1}{2}}$$

$$= \sqrt{\frac{2}{2}} = 1$$

Magnitude of a given vector is 1 then it is called unit vector.

9. Magnitude of $i + j$ is $\sqrt{1^2 + i^2}$
$= \sqrt{1+1} = \sqrt{2}$

10. According to phythogoras theorem 3



$s_1^2 + s_2^2 = h^2$

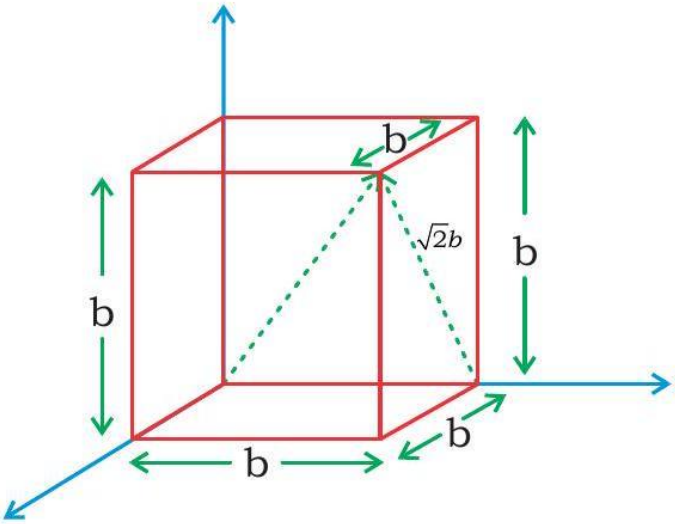
$(3)^2 + 4^2 = h^2$

$9 + 16 = h^2 \Rightarrow 25 = h^2$

$h = 5$ m

12. The unit vector along the diagonal passing through a origin of a cube is

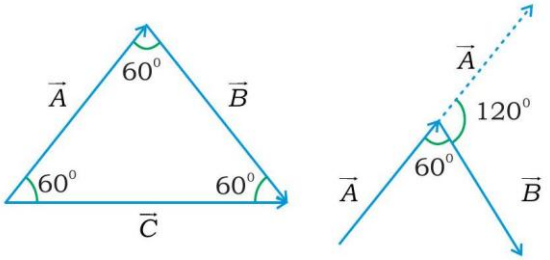
$T\left|\frac{b}{b}\right| = \frac{\$+\$+\hat{k}}{\sqrt{1^2+1^2+1}} = \frac{\$+\$+\hat{k}}{\sqrt{3}}$



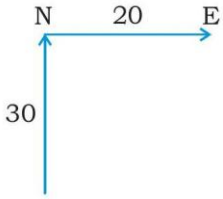
13. A vector of a magnitude z units along z-axis is $z\hat{k}$

A vector of a magnitude 12 units along z-axis is $12\hat{k}$

14. If head coincide with tail angle between two vectors is equal to external angle.



19.



15. $\overset{II}{S} = 30\$ + 20\$$

uI

$S = \sqrt{30^2 + 20^2}$

$= \sqrt{1300}$ m

21. Resultant direction is south west from initial direction.

Level – III

22)A,B,C 23) A,B,C,D 24) C 25) A 26) D

27) a-q,r,t;b-p,s;c-p,s;d-p,s 28) A 29) D 30) C

31) a-t;b-s;c-q;d-r 32) C

3. KINEMATICS

WORKSHEET-1

Level – I

1. B	2) B	3) A	4) D	5) B	6) C	7) C
8) D	9) A	10) A				

Level – II

1) A	2) B	3) B	4) A	5) D	6) B	7) A
8) B	9) A	10) B	11) D	12) A	13) B	14) B 15)
D	16) A	17) D	18) C	19) D	20) B	21) A

Level – III

22) D	23) A	24) B	25) A	26) a-q; b-s; c-p; d-r
27) A,B	28) A,B,D	29) B	30)A	31) A
32) a-p,s; b-t; c-q; d-r.				

WORKSHEET-2

Level – I

1) D	2) B	3) D	4) C	5) C	6) A	7) B
8) D	9) C	10) D	11) A			

1. Towards East = 4 m

   Towards North = 3n

   Towards Height = 12 m

   Total distance = 4 m + 3 m + 12 m = 19 m

2. $OA \rightarrow 8$ km

   $AB \rightarrow 4$ km

   $BC \rightarrow 3$ km

   $CD \rightarrow 1$ km

   $OA + AB + BC + CD = 8 \text{ km} + 4 \text{ km} + 3 \text{ km} + 1 \text{ km} = 16 \text{ m}$

3. Towards North = 6 m

   Towards East = 8 m

   Vertically upwards = 10 m

   Total distance = $6m + 8m + 10m = 24m$

4. To complete one revolution in circular path = $2\pi r$

   Half revolution = $\frac{2\pi r}{2} = \pi r$

5. To complete one revolution = $2\pi r$

   To complete 5 revolutions = $5 \times 2\pi r = 10\pi r$

   Distance = 25 m+= 25 m = 50 m

   Displacement = 0

6. Conceptual

7. Distance = AO + OA

   = 25 + 25 = 50 m

   displacement = 0

   = OA − OA = 25 − 25 = 0

9. Displacement = diameter

   = 2r

Level – II

1) D	2) D	3) A	4) B	5) B	6) B	7) C
8) C	9) C	10) C	11) D	12) C	13) B	14) A 15) D
16) A						

1. $AB^2 = OA^2 + OB^2 = (8)^2 + (6)^2 = 64 + 36 = 100$

   $AB = \sqrt{100} = 10$ m

2. Distance = OA + AO + OB

   = 25 m + 25 m + 25 m
   = 75 m

3. Distance moved through the book = 8 cm → Distance moved through the table = 6 cm ←

   ∴ Displacement = 8 − 6 = 2 cm

4. Distance = OA + AO + OB

   = (20 + 20 + 20)m

   = 60 m

   Distance

   $= OA - AO + OB$
   $= (20 - 20 - 20)$m = 20 m

   $OC^2 = OD^2 + DC^2$
   $= (20^2 + 20^2)m$
   $= (400 + 400)m$
   $OC^2 = 800m$
   $\therefore OC = \sqrt{800}m = (\sqrt{400 \times 2})m = 20\sqrt{2}m$

3. Distance = 8 cm $\overset{8\text{ cm}}{\longrightarrow}$ and then 6 cm displacement = 8 − 6 = 2 cm

4. Distance $= 50 + 40 + 20\ d = 110m$

Displacement $= 50$ cm

$OC^2 = (30)^2 + (40)^2 = 900 + 1600 = 2500$

$x = 50m$

7. Circumference of circle (distance) $= 2\pi$r

Half of its revolution $= \frac{2\pi r}{2} = \pi r$

distance $= \pi r$

radius $= \frac{\text{diameter}}{2}$

Displacement $= 2R$

8. $OC^2 = (20)^2 + (20)^2$

$OC^2 = 400 + 400$

$OC^2 = 800$

$OC = \sqrt{800} = \sqrt{400 \times 2}$

$= \sqrt{400} \times \sqrt{2}$

$OC = 20\sqrt{2}$

9. Towards East $= 3$ m

Towards North $= 4$ m

$OA^2 + AB^2 = OB^2$

$OB^2 = 25$

$OB = \sqrt{25} = 5$ m

10. 8 m due to East

6 m due to North

8 m due to West

Total distance $= 8m + 6m + 8m = 22m$

11. Above problem displacement 6 m towards South

12. Radius $= 700$ m

Displacement $= 2$R

$= 2 \times 700$ m
$= 1400$ m

13. $OC^2 = (20)^2 + (20)^2$

$OC^2 = 400 + 400$

$OC = \sqrt{800}$

$OC = \sqrt{400} \times \sqrt{2}$

$OC = 20\sqrt{2}$ cm

14. Displacement (x) = 55 m − 50 m = 5 m

t = 5sec

$x = 5m$

15. $2\pi r$

$2\pi r \times 2 = 4\pi r$

16. Total Distance = 10 km + 5 km = 15 km

Level – III

17) B 18) C 19) B 20) A,B 21) A-Q,B-P,C-S,D-P

22) B 23) A 24) A 25) B 26) B 27) A,D

28) A-Q,B-P,C-R,D-S 29) C 30) B 31) B

WORKSHEET-3

Level – I

1) C 2) C 3) B 4) D 5) C 6) A 7) A

8) B 9) A 10) B

Level – II

1) C 2) A 3) 4 4) C 5) B 6) A 7) B

8) B 9) C 10) C 11) A 12) B,D 13) C 14) B

15) C 16) A 17) A 18) C 19) A 20) A 21) C 22) D 23) A

1. Average speed $= \frac{\text{Total distance covered by}}{\text{total time taken}}$

$= \frac{200 \text{ m}}{14} = 8$ m/s

2. Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$2.5 \text{ m/s} == \frac{\text{Total distance}}{1.5\text{hr}}$

2.5 m/s canverts to km/hr

$2.5 \times \frac{18}{5} = 0.5 \times 18 = 9$

9 km/hr $= \frac{\text{Total distance}}{1.5\text{hr}}$ covered body

Total distance covered body $= 9 \times 1.5$

= 13.5 km

3. Distance covered at the start of the trip = 22687 km

Distance covered at the end of the trip = 22791 km

Distance covered on the journey = 22791 − 22687 = 104 km = 104000 m

Time taken for entire journey = 4 hours.

$= 4 \times 3600 = 14400$sec

Average speed in kmph $= \frac{\text{distance}}{\text{time}} = \frac{104}{4} = 26$kmph

Average speed in $ms^{-1} = \frac{1,04,000}{14,400} = 7.22\ \text{ms}^{-1}$

4. In order to cross the bridge completely. Then it must cover distance = train largest bridge length

Train length = 100 m

Bridge length = 0.5 km = 500 m

Train + Bridge lengths $= 100m + 500m = 600m$

Speed $= \frac{\text{distance}}{\text{T}}$

Speed = 72kmph $= 72 \times \frac{5}{18} = 20$ m/s

20 m/s $= \frac{600\ \text{m}}{\text{time}} \Rightarrow \text{Time} = \frac{600\ \text{m}}{20\ \text{m/s}} = 30$se

Time = 30sec.

5. Velocity $= \frac{\text{Displacement}}{\text{Time}}$

Here, displacement is equal to 2 m and time is 1 second.

So velocity $= \frac{2\ \text{m}}{1\text{sec}} = 2$ m/s

6. Average speed $= \frac{\text{total distance}}{\text{total time}}$

Total time = 1 minute (60 s) + 2 minute (120sec)

= 3 minutes or 180 seconds

Total distance = (Speed × Time) + (speed + time)

= (1 m/s × 60 s) + (3 m/s × 60sec)

$= 60m + 180 = 240m$

Average speed $= \frac{240\ \text{m}}{120\ \text{s}} = 2$ m/s

7. $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2\times2.5\times4}{6.5} = \frac{40}{13}$

8. In order to cross the bridge completely then it must cover distance = train length

- bridge length

Train length = 200 m

Bridge lengths = 600 m

Train + Bridge length = 200 + 600 = 800 m

Speed $= \frac{\text{distance}}{\text{time}}$

Speed $= 36\text{kmphr} = 36 \times \frac{5}{18}\ \text{m/s} = 10\ \text{m/s}$

Speed $= \frac{\text{distance}}{\text{time}}$

$10m/s = 600 + 200 = \frac{800m}{\text{Time}}$

Time $= \frac{800}{10} = 80\text{sec}$

9. $v = 3\left(\frac{1\text{ cm}}{\text{year}}\right) \times \left(\frac{1\text{ m}}{100\text{ cm}}\right)$

$= 0.03(\text{ m/ year })$

First we convert the unit of speed from cm/ year to m/ year by dividing it by 100 . since 1 m = 100 cm

9. $v = \frac{x}{t} \Rightarrow t = \frac{x}{v}$

t = 1500 m$t = \frac{1500(\text{ m})}{0.03\left(\frac{\text{m}}{\text{year}}\right)}$

$t = 50.000$ year.

10. After each circular lap. The car reaches its initial position. There fore displacement is 0 . This implies average velocity =

Average velocity $= \frac{\text{total displacement}}{\text{Total time}} = 0$

Average speed $= \frac{\text{Total distance}}{\text{total time}}$

Total distance $= 2\pi$r

$= 2 \times 3.14 \times 100$

Average speed $= \frac{2\times3.14\times100}{62.8}$

11. Let the distance between A and B we ' d ' we are from use the formula, distance = speed × time

Time taken to go from A to B is $t_1 = \frac{d}{v_1}$

Time taken to go from B to A is $t_2 = \frac{d}{v_2}$

Average speed $= \frac{\text{total distance}}{\text{total time}} = \frac{2d}{(t_1+t_2)}$

$= \frac{2d}{\frac{d}{v_1} + \frac{d}{v_2}} = \frac{2(v_1v_2)}{v_1 + v_2}$

12. $\frac{2}{5}$ sec with time $t_1 = \frac{\frac{2}{5}\text{se}}{v_1} = \frac{2s}{5v_1}$

$\frac{3}{5}$sec with time $t_2 = \frac{\frac{3}{2}se}{v_2} = \frac{3se}{5v_2}$

(speed) $= \frac{d}{T} = \frac{d}{t_1+t_2}$

$= \frac{d}{\frac{2s}{5v_1} + \frac{3\text{sec}}{5v_2}} = \frac{5}{\frac{2v_2+3v_1}{v_1v_2}} = \frac{5v_1v_2}{2v_2+3v_1}$

13. Average speed $= \frac{\text{total distance}}{\text{Total time taken}}$

First we will calculate the total distance covered $= \frac{\text{d}}{3} + \frac{d}{3} + \frac{d}{3} = 3\left(\frac{d}{3}\right) = d$

Time taken $= \frac{\text{distance}}{\text{speed}}$

$\Rightarrow \text{t}_1 + t_2 + t_3 = \frac{\frac{d}{3}}{v} + \frac{\frac{d}{3}}{2v} + \frac{\frac{d}{3}}{3v} = \frac{d}{3v}\left(\frac{11}{6}\right) = \frac{11d}{18} \;\; \therefore V_{\text{avg}} = \frac{d}{\frac{11d}{18v}} = \frac{18v}{11}$

14. $V_{\text{avg}} = \frac{2V_1V_2}{V_1+V_2}$

$40 = \frac{2(V)60}{60+V}$

$V = 30\text{kmph}$

15. $t_1 = \frac{d}{40}$

$t_2 = \frac{d}{60}\; t_3 = \frac{d}{60}$

usage $(\text{A} - \text{C}) = \frac{\text{Tota distance}}{\text{total time}} = \frac{2\text{ d}}{\text{t}_1+t_2} = \frac{2d}{\frac{d}{40}+\frac{d}{60}} = \frac{2\times40\times60\times d}{100\text{ d}} = 48\text{kmph}$

Usage $(A - D) = \frac{3d}{\frac{d}{40}+\frac{d}{60}+\frac{d}{60}} \Rightarrow \frac{3d}{\frac{d}{40}+\frac{d}{30}} = \frac{40\times30\times3d}{70d} = \frac{360}{7}\text{kmph} = 51.4\text{kmph}$

16. Length of the train be x meters and its speed by ym/sec.

Then $\frac{x}{y} = 15 \Rightarrow y = \frac{x}{15}$

$\therefore \frac{x+450}{45} = \frac{x}{15}$

$\Rightarrow 15x + 15 \times 450 = 45x$

$\Rightarrow 15 \times 450 = 30x$

$\Rightarrow \frac{15\times450}{30} = 225\text{ m}$

17. In first case

$\therefore$ distance (d) = 60 km

Speed = 120kmh

Time taken $= \frac{\text{distance}}{\text{speed}} = \frac{60}{120} = \frac{1}{2}hr = \frac{1}{2}\times60 = 30\text{ min.}$

In second case

Distance = 60 km

Speed = 80kmph

Time taken = $\frac{\text{distance}}{\text{speed}} = \frac{60}{80} = \frac{3}{4}\text{hr} = \frac{3}{4} \times 60 = 45$ min

∴ Total time taken (t) = $t_1 + t_2 = \frac{1}{2} + \frac{3}{4} = \frac{5}{4} = \frac{5}{4} \times 60 = 75$ min.

18. Let total journey be x

Time with speed $v = \frac{\left(\frac{1}{2} \times x\right)}{v} = \frac{x}{2v}$

Remaining part of journey = $\frac{1}{2 \times x}$

Time with speed $2v = \frac{\frac{1}{4} \times x}{2v} = \frac{x}{8v}$

Time with speed $4v = \frac{\frac{1}{4}x}{4v} = \frac{x}{16v}$

Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

Speed = $\frac{x}{\frac{x}{2v} + \frac{x}{8v} + \frac{x}{16v}}$

Average speed = $\frac{x}{\frac{11x}{16v}} = \frac{16v}{11}$

19. Let us Assume that the particle travelled from A to B Where $AO = OB$. Let the total distance covered by the particle be d.

∴ Half of distance = $\frac{d}{2}$

Let us divide the second half distance into two parts d_3 and d_4

∴ $d_3 = v_3 \times t_3$ and $d_4 = v_4 \times t_4$

$d_3 = 4.5 \times t_3$ and $d_4 = 7.5 \times t_3$ Q$[t_3 = t_4]$

∴ Average speed for second half distance

$v_2 = \frac{d_3 + d_4}{t_3 + t_4} = \frac{4.5t_3 + 7.5t_3}{2 + 3}$ $[\text{Q}t_3 = t_4]$

$= \frac{12 + 3}{2 + 3} = 6$ m/s → (1)

Average speed for the entire journey

$v = \frac{d_1 + d_2}{t_1 + t_2} = \frac{d}{\frac{d_1}{v_1} + \frac{d_2}{v_2}}$ But $d_1 = d_2 = \frac{d}{2}$

$\therefore v = \frac{d}{d\left(\frac{1}{2v_1} + \frac{1}{2v_2}\right)} = \frac{2v_1v_2}{v_1 + v_2} = \frac{2 \times 3 \times 6}{3 + 6} = \frac{36}{9} = 4$ m/s

20. $v = 720$ m/s

$d = 3240m$

$t = \frac{3240\ \text{m}}{720\ \text{m/s}} = 4.5\text{sec}$

21. distance = Speed × time

$= 240 \times 5 = 1200\text{Km}$

Now we have to speed over $1 + \frac{2}{3}$ hours $= \frac{5}{3}$ hours

$\text{speed} = \frac{\text{dis tance}}{\text{time}}$

$\text{speed} = 1200 \times \frac{5}{3}\text{kmph} = \frac{1200\times3}{5} = 720\text{kmph}$

22. Let the required distance be x km

Then $\frac{x}{3} + \frac{x}{2} = 5$

$\frac{m + 3x}{6} = 5$

$5x = 6 \times 5$

$x = \frac{30}{5} = 6$ km

23. Let the required distance be km.

Difference in the time taken at two speeds

$= 12\text{min} = \frac{1}{5}hr \Rightarrow \frac{x}{5} - \frac{x}{6} = \frac{1}{5} \Rightarrow x = 6$ km

Level – III

24) A,B,C 25) D 26) A 27) D 28) a-q; b-r,s; c-p; d-s

29) 6m/s 30) a-q; b-p; c-s; d-r 31) A,B

25. $\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{30\ \text{m}}{30\text{sec}} = 1\ \text{m/s}$

26. $\text{speed} = \frac{\text{distance}}{\text{time}}$

$\text{time} = \frac{\text{dis tance}}{\text{speed}} = \frac{15\ \text{km}}{120/\text{hr}} = 0.125\text{hr}$

27. distance = speed × time = 15 × 45 = 675 m

28. a) $q, b)r, c)p, d)s$

29. $\text{Speed} = \frac{\text{distance}}{\text{speed}} = \frac{60\ \text{m}}{10\ \text{s}} = 6\ \text{m/s}$

WORKSHEET-4

Level – I

1) B 2) A 3) A 4) B 5) D 6) B 7) D

8) C 9) C 10) C

Level – II

7) C	8) B	9) C	10) B	11) A	12) D	13) C
14) A	15) D	16) C	17) C	18) B	19) B	20) D
21) A	22) A					

1. average velocity $= \frac{\text{total displacement}}{\text{total time}} = \frac{120\text{ m}+120\text{ m}}{60\text{ s}} = \frac{240}{60}\text{ m/s} = 4\text{ ms}^{-1}$
2. velocity $= \frac{\text{total displacement}}{\text{time}} \Rightarrow \text{time} = \frac{\text{total displacement}}{\text{velocity}}$

   $\Rightarrow \text{time} = \frac{1.5\times10^8}{3\times10^5} = \frac{15\times10\times10^2}{3\times10} = 500\text{ s}$
3. $v = \frac{2V_1V_2}{V_1+V_2}$
4. Average speed $= \frac{2v_1v_2}{v_1+v_2} = \frac{2\times20\times80}{20+80} = -$

   $= 32\text{kmph}$
5. average velocity $V_{\text{avg}} = \frac{V_1+V_2}{2}$
6. s/2 + s/3
7. Average velocity $= \frac{\text{total displacement}}{\text{total time}}$

   since the average velocity is zero

   Therefore, the total displacement is zero

   Therefore, the car returns in its original position

   Thus, Distance covered by the car with velocity 20 m/s in 5 seconds = distance covered by the car with velocity v in 10 seconds

   using,

   dis tance = speed × time

   $20 \times 5 = v \times 10$

   $v = 10\text{ m/s} \Rightarrow 10\text{ ms}^{-1}$ due west
8. $(40-20)\text{m/s} = 20\text{ m/s}$
9. -200kmph

   Change in velocity $= V_1 + V_2$ due east
10. $[10\text{ m/s} - 5\text{ m/s}]$ due east $= 5\text{ m/s}$ east
11. Average speed $= \frac{2V_1V_2}{V_1V_2} = \frac{2\times40\times60}{40+60} = \frac{2\times2400}{100} = 48\text{kmph}$

    Average velocity $= 0$
12. equal (or) less than one.
13. Average velocity $= \frac{V_1+V_2}{2}$

14. time taken $= \frac{\text{distance}}{\text{speed}} \Rightarrow \frac{60\text{ km}}{\left(\frac{80+40}{2}\right)}$kmph < speed $= \frac{V_1+V_2}{2}$

$$t = \frac{60}{(120/2)} = 60/60 = 1hr$$

Average velocity $= \frac{V_1+V_2}{2} = \frac{80+40}{2} = \frac{120}{2} = 60$kmph

15. Given that bike messenger travel 300 m north 500 m east, 500 m south and finally 100 m west

Resultant displacement, $s = \sqrt{400^2 + 200^2}$

$$= \sqrt{160000 + 40000}$$
$$= \sqrt{200000}$$
$$= \sqrt{20 \times 10^4} = \sqrt{10^4 \times 2^2 \times 5} = 10^2 \times 2 \times \sqrt{5}$$
$$= 100 \times 2 \times \sqrt{5}$$
$$= 200\sqrt{5}\text{ m}$$

average velocity $= \frac{\text{resul tan } t \text{ displacement}}{\text{total time taken}}$

$$= \frac{24200\sqrt{5}}{45_9 \times 60_3} = \frac{2\sqrt{5}}{27} = \frac{4.472}{27}\text{ m/s}$$

16. Average velocity $= \frac{5}{150\times60} = \frac{1}{18} \times 10^{-2} = 5.6 \times 10^{-3}\text{cms}^{-1}$

17. time $= \frac{\text{total displacement}}{\text{uniformvelocity}} = \frac{1100}{45\times5/18} = \frac{1100}{12.5} = 88$sec

18. Average velocity $v = \frac{3abc}{ab+bc+ca}$

$$v = [1/3[1/a + 1/\text{b} + 1/\text{c}]] = 1 \Rightarrow \text{v} = \frac{3abc}{ab + bc + ca}$$

19. time $t = \frac{500}{[18\text{kmph}+72\text{kmph}]}$

$$t = \frac{500}{90\text{kmph}} = \frac{500}{590 \times 5/18} = \frac{500}{25} = 20\text{sec}$$

$\therefore$ time t = 20sec

21. The magnitude of her average velocity is zero.

22. total distance = x

time taken for $1/_3^{rd}$ distance $= \frac{1\frac{x}{3}}{v_1} = \frac{x}{3v_1}$

Remaining distance is covered with a velocity v_2 for half of the time and with a velocity

Average velocity for $v_2 \& v_3 = \frac{v_2+v_3}{2}$

time taken for remaining distance $\frac{3}{\frac{v_2+v_3}{2}} 3(v_2 + v_3)$

$$= \frac{2x}{\frac{3}{\frac{v_2+v_3}{2}}} = \frac{4x}{3(v_2+v_3)}$$

∴ Average velocity of the scooter for the journey is $= \frac{x}{\frac{x}{3v_1}+\frac{4x}{3(v_2+v_3)}}$

$$= \frac{x}{\frac{3(v_2+v_3)x+4x(3v_1)}{3v_1(3)(v_2+v_3)}}$$

$$= \frac{x}{\frac{3x[4v_1+v_2+v_3]}{9v_1[v_2+v_3]}}$$

∴ Average velocity of scooter for wholes average $= \frac{3v_1[v_2+v_3]}{4v_1+v_2+v_3}$

Level – III

23) C 24) B 25) A 26) a-p,r; b-q, r; c-p,s; d-q,s

27) A,B,C 28) A,B 29) A 30) C 31) A

32) a-q, b-s, c-p, d-c

23, 24, 25.

Given that

Skilled driver has managed to drive his car with a uniform velocity of 5 m/s for 10 minutes

i.e, 10 minutes = 10 × 60 = 600 seconds

⇒ Driver drive his car with uniform velocity of 5 m/s for about 600 seconds.

⇒ Average velocity of car in first 5 minutes = 5 m/s

⇒ Velocity of car at the end of 5th minute = 5 m/s

⇒ Instantaneous speed of car at t = 500sec in also 5 m/s

29. Given that,

radius of circle = 1 m

It reaches to its initial position in 10 s

i.e, displacement = 0

∴ velocity = 0

30. Distance = 500 km

time = 1hr

velocity $= \frac{d}{t} = \frac{500}{1} = 500$ km/hr

31. If motor cucle covers half of total path,

displacement = 2r

time = t

∴ velocity = $\frac{2r}{t}$

WORKSHEET-5

Level – I

1) C	2) A	3) A	4) B	5) B	6) A	7) B
8) B	9) C	10) D				

Level – II

1) A	2) B	3) A	4) D	5) A	6) B	7) B
8) A	9) C	10) C	11) A	12) B	13) D	14) A 15)
B	16) D	17) D	18) A	19) B	20) D	

4. Initial velocity of a bus $u = 12$ m/s

final velocity of a bus $v = 6$ m/s

time ' t ' = 35

average acceleration = $\frac{(v-u)}{t} = \frac{6-12}{3} = \frac{-6}{3} = -2$ m/s^2

7. Acceleration $a = 4$ ms^{-2}

velocity $v = 72$ ms^{-1}

t = ?

formula $t = a = \frac{u-v}{t}$

$\Rightarrow t = \frac{v-u}{a}\frac{72-0}{4} = \frac{72}{4} = 18$sec

8. $u = 10$ ms^{-1}

$v = 20$ ms^{-1}

$t = 2$sec

a = ?

formula acceleration of the body

$a = \frac{v-u}{t} = \frac{20-10}{2} = \frac{1\sigma^5}{}$

$a = 5$ m/s^2

9. Initial velocity $4 = 10$ ms^{-1}

time $t = 3$sec

final velocity v = 0

to calculate average deceleration

average duration = $\frac{v-u}{t} = \frac{0-60}{3} = \frac{-60}{3} = -20$ m/s^{-2}

duratoin = 20 m/s

10. $v = 36\text{kmph} \Rightarrow 36 \times \frac{5}{18} = 10$

$t = 10\text{sec}$

$u = 6$ m/s

$a = ?$

$$a = \frac{(v-u)}{t} = \frac{10-10}{10} = \frac{10}{10} = 1$$

$a = 1$ m/s^2

11. $u = 5ms^{-1}$

$v = 10$ ms^{-1}

s = 50 m

$a = ?$

formula

$v^2 - u^2 = 2as$

$10^2 - 5^2 = 2 \times a \times 50$

$$\frac{100-25}{100} = a$$

$$\frac{100-25}{100} = a$$

12. Average velocity = 0.34 m/s

Displacement = 3.06 m

Change velocity (v – u) = 0.18 ms^{-1}

$v_{\text{avg}} = \frac{s}{t}$

$$0.34 = \frac{3.34}{t}$$

$$t = \frac{3.06}{0.34} = \frac{3.06}{0.34} = 9 \text{ s}$$

Acceleration $a = \frac{v-u}{t} = \frac{0.18}{a} = 0.02$ m/s^{-2}

13. Initial velocity 4 = 10 ms^{-1}

Final velocity $v = 30$ ms^{-1}

time $t = 5$sec

formula

acceleration is given by

$$a = \frac{v-u}{t} = \frac{30-10}{5}$$

$= \frac{2\sigma^4}{b_1} = 4 \text{ m/s}^2$

The initial velocity $u = 3$ m/s

final velocity $v = 60$ m/s

$a = \frac{v-u}{t}$

$t = \frac{v-u}{a} = \frac{60-30}{4}$

$= \frac{30}{4} = 7.5$ s

$t = 7.5$sec

15. $u = 0m/s$ A to B

$v = v_{\max}$

$a = \infty$

$v = u + at$

$v_{\max} = 0 + \propto t_1$

$t_1 = \frac{v_{\max}}{\propto}$

B to C

$u = v_{\max}$

$v = 0$ m/s

$a = -\beta$

$v = u + at_2$

$0 = v_{\max} = \beta t_2$

$t_2 = \frac{v_{\max}}{\beta}$

$t = t_1 + t_2$

$t = v_{\max}\left[\frac{1}{\propto} + \frac{1}{\beta}\right]$

$\frac{1}{\left[\frac{1}{\propto} + \frac{1}{\beta}\right]} = v_{\max}$

$t\frac{\propto \beta}{(\propto + \beta)} = v_{\max}$

16. $u = 0$ m/s

$v = m/s$

$t = t_1, a = \propto$

$v = u + \propto t_1$

$t_1 = \frac{v_{max}}{a}$

B to C

$u = v_{max}$

$t = t_2, a = -\beta$

$v = u + at$

$= v_{max} - \beta t_2$

$t_2 = \frac{v_{max}}{\beta}$

$\frac{t_1}{t_2} = \frac{\frac{v_{max}}{\propto}}{\frac{v_{max}}{\beta}} = \frac{v_{max}}{\propto} \times \frac{v_{max}}{v/\ max}$

$\frac{t_1}{t_2} = \frac{\beta}{\propto}$

Level – III

21) B,D 22) 60 23) B 24) B 25) 5

26) a-p; b-q; c-r; d-s 27) D 28) 5 29) A 30) B 31) C

22. Acceleration $a = 4ms^{-1}$

timet $= -15s$

$u = 0$

formula

$a = \frac{(v-u)}{t}$

$4 = \frac{v-u}{t}$

$4 \times 15 = v$

$v = 60$ m/s

23. Initial velocity $u = 18$ km/hr

final velocity $v = 72$ km/hr

$a =?$

$t = 30$ s

$a = \frac{v-u}{t} = \frac{72-18}{\frac{30}{3600}}$

$= \frac{54 \times 3600}{30}$

$= 54 \times 120\ a = 6480$ km/hr^2

24. Change in velocity $(v-u) = 54$ km/hr

$t = 1\text{min} = \frac{1}{60} hr$

acceleration $a = \frac{(v-u)}{t} = \frac{54/1}{1/60}$

$= 54 \times 60$

$a = 3240 \text{ km/hr}^2$

25. Initial velocity $a = 1.8$ km/hr$108 \times \frac{5}{18} = 6 \times 5 = 30$ m/s

Final velocity

$= 36$ km/hr $= 36 \times \frac{5}{18} \pm 2 \times 5 = 10$ m/s

$t = 4$ s

formula

acceleration $\frac{v-u}{t} = \frac{10-30}{4} = \frac{-20}{4}$

average $4 - (-5) \ -5m/s^2$

duratoin $= +5 \text{ m/s}^2$

26. Initial velocity $4 = 60$ m/s

final velocity $v = 80$ m/s

$t = 4$ s

$a = ?$

formula

$a = \frac{v-u}{t}$

$= \frac{80-60}{40} = \frac{20}{4}$

$a = 5 \text{ m/s}^2$

WORKSHEET-6

Level – I

1) A	2) B	3) A	4) C	5) D	6) C	7) C
8) A	9) C	10) A	11) B			

Level – II

1) B	2) C	3) B	4) D	5) A	6) C	7) D
8) B	9) C	10) D	11) D	12) B	13) B	14) A 15)
C	16) C	17) C	18) C			

2. $v = 5 + 6t$

$v = 35$ m/s

$t = ?$

Use $v = u + at$

$35 = 5 + 6t$

$t = 5\text{sec}$

3. $u = 7$ m/s, $a = 2$ m/s^2, $s = ?$, $t = 2\text{sec}$

$s = ut + \frac{1}{2} \cdot at^2$

$s = 7 \cdot 2 + \frac{1}{} \cdot$

$s = 18m$

4. $u = 0$, $a = 3m/s^2$, $t = 4$, $s = ?$

$s = ut + \frac{1}{2}at^2$

$s = 0 \times 4 + \frac{1}{2} \cdot 3 \cdot 4^2 = 24m$

5) $u = 20$ m/s, $v = 0$ m/s, s $= 200$, $a = ?$

$v^2 - u^2 = 2as$

$0^2 - 20^2 = 2.a.200$

$-400 = 400a$

$a = -1$ m/s^2

decelaration $a = 1$ m/s^2

6. $u = 0, v = 10$ m/s, $t = 15.$s, $a = ?$

$a = \frac{v - u}{t} = \frac{10 - 0}{15} = \frac{10}{15} = 0.67$ m/s^2

7. $a = 2$ m/s^2, $t = 10\text{sec}, \text{s} = ?$

Suppose $u = 0$

$s = ut + \frac{1}{2}at^2$

$= 0 \times 10 + \frac{1}{2} \cdot 2 \cdot 10^2$

$s = 100m$

8. $a = 3.20$ m/s^2, $t = 32.8\text{sec}, v = 0$, s $= ?$

$-3.20 = \frac{0 - u}{32.8}$

$u = 104.96$ m/s

$v^2 - u^2 = 2as$

$0^2 - (105)^2 = 2 \cdot (-3.2)s$

$-11{,}025 = -6.4.s$

$S = 1.721.656m$

$v^2 - u^2 = 2as$

$0^2 - (104.96)^2 = 2 \cdot (-3.2)s$

$s = \dfrac{11016.6}{6.4}$

$s = 1721.34$ m

9. $u = 8m/s, a = 2m/s^2, s = 9m, v = ?$

$v^2 - u^2 = 2as$
$v^2 - 8^2 = 2.2.9$
$v^2 = 64 + 36$
$v = 10$ m/s

10. $u = 10$ m/s, $a = 4$ m/s^2, $t = 5$sec, $v = ?$

$v = u + at$

$v = 10 + 4 \times 5$

$v = 30$ m/s

11. $u = 35$ m/s, $s = 10$ cm, $v = 0, a = ?$

$v^2 - u^2 = 2as$

$0^2 - (35)^2 = 2(a) \cdot 10 \times 10^{-2}$

$-1225 = a(20 \times 10^{-2})$

$a = \dfrac{-1225}{20} \times {}^{5}100 = 6125$ m/s^2

12. $u = 36$kmph, $a = 2$ m/s^2, $t = 3$sec, s = ? $u = 10$ m/s

$\because$ 36kmph $= 36 \times \dfrac{5}{10} = 10$ m/s

$s = ut + \dfrac{1}{2}at^2$

$s = 10 \times 3 + \dfrac{1}{} \cdot 2 \cdot 3^2 = 39m$

13. $u = 0$ m/s, $v = 36$kmp, $h = 10$ m/s,

$t = 12$sec

$a = \dfrac{v - u}{t} = \dfrac{10 - 10}{12} = \dfrac{5}{6}$ m/s^2

14. $S_p = ut + \frac{1}{2}at^2$

$= (20 \times 50) + \dfrac{1}{2} \times 1 \times 50^2$

$= 2250$ m

$S_q = ut = 20 \times 50 = 1000$ m $S_p - S_q = 2250 - 1000 = 1250$ m

$x + 800 = 1250$

$x = 1250 - 800 = 450$ m

15. $v^2 - u^2 = 2as$

$0^2 - 20^2 = 2(-5) \times d_2$

$d_2 = 40m$

$d_1 + d_2 = 50$

$t_0 = 0.55\text{sec}$

16. $S_A = ut + \frac{1}{2}at^2$

$S = 0 + \frac{1}{2} \times 8 \times t^2 \rightarrow (1)$

$S_B = ut = 16t \rightarrow (2)$

$S_A = S_B\ t = 4\text{sec}\ d = 64$ m

17. $v_2^2 - v_1^2 = 2as$

$a = \frac{v_2^2 - v_1^2}{2s}$

$v_2^2 - v_1^2 = 2 \cdot a \cdot \left(\frac{s}{2}\right)$

$v^2 = v_1^2 +$

$v^2 = v_1^2 + \frac{v_2^2 - v_1^2}{2}$

$v^2 = \frac{2v_1^2 + v_2^2 - v_1^2}{2}$

$v^2 = \frac{v_2^2 + v_1^2}{2}$

$v = \sqrt{\frac{v_2^2 + v_1^2}{2}}$

18. $u = 0\ a = a$

$s_n = u + \frac{a}{2}(2n - 1)$

$s_1 = 0 + \frac{a}{2}(2(1) - 1)$

$s_1 = \frac{a}{2}$

$s_2 = 0 + \frac{a}{2}(2(2) - 1)\ n = 2$

$= \frac{a}{2}(3)$

$s_3 = 0 + \frac{a}{2}(2(3) - 1)\ n = 3$

$= \frac{a}{2}(5)$

ratio $s_1 : s_2 : s_3$

$\frac{a}{2} : \frac{3a}{2} : \frac{5a}{2} = 1 : 3 : 5$

9-22: Conceptual

Level – III

19) A,B,C 20) A 21) A,B,C,D 22) C 23) D 24) A 25) B 26) a-p,r,s;b-p;c-p,q;d-p,t

23. $u = 0, a = 3\ \text{m/s}^2, t = 3\text{sec}$

$v = u + at$

$v = 0 + 3 \times 3$

$v = 9\ \text{m/s}$

24.

$t = 3\text{sec}\ s = ?$

$s = ut + \frac{1}{2}at^2$

$s = 0 \times 3 + \frac{1}{2}(3) \cdot 3^2$

$s = 13.5\ \text{m}$

25. $v^2 - u^2 = 2as$

$v^2 - 0^2 = 2.3.6$

$v^2 = 6^2$

$v = 6\ \text{m/s}$

$a \to p, r, s, b \to p, c \to p, q, d \to p, t$

WORKSHEET-7

Level – I

1) D 2) D 3) D 4) B 5) C 6) C 7) C

Level – II

1) D 2) B 3) A 4) C 5) A 6) D 7) A 8) B 9) D 10) B 11) A 12) C 13) D 14) C 15) C 16) A 17) A 18) B 19) A 20) B 21) B 22) A 23) C 24) B 25) B

1. $S_n = u + \frac{a}{2}(2n - 1)$

$\text{u} = 5\ \text{m/s}$

2. $S_n = u + \frac{a}{2}(2n-1)$

$S_n = 3 + 8(2n-1)$

$= 3 + \frac{16}{2}(2n-1)$

$a = 16\ \text{m/s}^2$

3. $S_n = u + \frac{a}{2}(2n-1)$

$150 = o + \frac{a}{2}(2 \times 8 - 1)$

$300 = a(15)$

$a = \frac{300}{15} = 20\ \text{m/s}^2$

4. $v = u + at$

$a = \frac{27.5 - 0}{10} = 2.75\ \text{m/s}^2$

$S_{10} = \frac{1}{2}at^2 = \frac{1}{2} \times 2.75 \times 100 = 2.75 \times 50$

$S_{20} = \frac{1}{2}at^2 = 2.75 \times 200$

$S_{20} - S_{10} = 2.75(200 - 50)$

$= 2.75 \times 150 = 412.5\ \text{m}$

5. $S_n = u + \frac{a}{2}(2n-1)$

$= 3 + \frac{2}{2}(5 \times 2 - 1)$

$= 3 + 9 = 12\ \text{m}$

6. $v^2 - u^2 = 2as$

$\frac{S_2}{S_1} = \left(\frac{u_2}{u_1}\right)^2 = \left(\frac{120}{60}\right)^2 = 4$

$S_2 = 4\ S_1 = 4 \times 20 = 80\ \text{m}$

7. Retardation between A& B = Retardation between B × C

$\frac{\left(\frac{v}{2}\right)^2 - v^2}{2 \times 3} = \frac{o^2 - \left(\frac{v}{2}\right)^2}{2x}$

$\frac{1}{3}\left(\frac{v^2}{4} - v^2\right) = \frac{1}{x}\left[0 - \frac{v^2}{4}\right]$

$-\frac{1}{3}\left[\frac{3v^2}{4}\right] = -\frac{v^2}{4} \cdot x$

$x = 1\ \text{cm}$

8. $S_1 = \frac{1}{2}a \times 10^2 = 50a$

$S_1 + S_2 = \frac{1}{2}a \times 20^2 = 200a$

$S_2 = 150a = 3S_1$

$S_2 = 3S_1$

9. $S = ut + \frac{1}{2}at^2$

For bus $S = \frac{1}{2} \times 5 \times t^2$

For car S = 50t

$\frac{5}{2}t^2 = 50t$

t = 20sec

v = u + at

$v = 5 \times 20 = 100$ m/s.

10. $u = 30$ m/s, $v = 10$ m/s, $t = 5$sec

$a = \frac{v-u}{t} = \frac{10-30}{5} = -4$ m/s^2

$S_n = u + \frac{a}{2}(2n-1), t = 3\text{sec}$

$= 30 + \frac{-4}{2}(5)$

$= 20$ m

11. $\frac{S_{nth}}{S_n} = \frac{9}{25}$

$\frac{u + a\left(t - \frac{1}{2}\right)}{ut + \frac{1}{2}at^2} = \frac{9}{25}$

$9t^2 = (2t-1)25$

$9t^2 - 50t + 25 = 0$

$t = 5/9\text{sec}\, t = 5\text{sec}$

12. $\frac{1}{t_1} = \frac{o+v}{2}$

$2 = vt_1$

uniform motion $3 = vt_2$

retarded motion $\frac{2}{t_3} = \frac{v+o}{2} \Rightarrow 4 = vt_3$

$a = v(t_1 + t_2 + t_3)$

$v = \dfrac{9}{\dfrac{10}{60}} = 54\text{kmph}$

13. $V^2 = 2a_1s_1 = 2a_2s_2$

$S_1 = \frac{v^2}{2a_1} = \frac{v^2}{10}$ and $S_2 = \frac{V^2}{2a_2} = \frac{V^2}{20}$

$\mathrm{S_1 + S_2 = 1500}$

$\dfrac{V^2}{10} + \dfrac{V^2}{20} = 1500$

$\mathrm{V = 100\ m/s}$

$V = a_1t_1 = a_2t_2$

$100 = 5t_1 = 10t_2$

$\mathrm{t_1 = 20sec, t_2 = 10sec}$

$\mathrm{t_1 + t_2 = 10 + 20 = 30sec}$

14. $\mathrm{u = 50\ km/hr, V = 0, S = 6\ m}$

$a = \dfrac{V^2 - u^2}{2s} = \dfrac{-u^2}{2a} = \dfrac{-\left(\frac{500}{36}\right)}{2 \times 6} = -16\ \mathrm{m/s^2}$

$u = 100\ \mathrm{km/hr} = 100 \times \dfrac{5}{18} = 27.78\ \mathrm{m/s}$

$$S = \frac{V^2 - u^2}{2a} = \frac{-u^2}{2a} = \frac{-(27.78)^2}{-2 \times 16} = 24m$$

15. $u - \frac{u}{20} = \frac{19u}{20}$

$V^2 - u^2 = 2as$

$\left(\dfrac{19u}{20}\right)^2 - u^2 = 2ax$

$\dfrac{-39}{400}u^2 = 2ax$

$0 - u^2 = 2anx$

$n = -\dfrac{u^2}{2ax} = \dfrac{400}{3a} = 10.3$

$\mathrm{n = 11}$

16. $\frac{S}{S_n} = \frac{n^2}{2n-1}$

$\dfrac{S}{S_3} = \dfrac{3^2}{2 \times 3 - 1} = \dfrac{9}{6-1} = \dfrac{9}{5}$

17. $S_n = u + \frac{a}{2}(2n - 1)$

$10 = u + \frac{a}{2}(2 \times 3 - 1)$

$16 = u + \frac{a}{2}(2 \times 4 - 1)$

Solve equation-(1) & equation-(2) u = −5 m/s

18. $v = 1 \times t_1$

$S_1 = \frac{1}{2} \times 1 \times t_1^2 = \frac{t_1^2}{2} - (2)$

$v = 3t_{2-}(3)$

$S_2 = ut_2 - \frac{1}{2}3 \times t_2^2 = t_1 t_2 - \frac{3}{2}t_2^2$

from (1) & (3) $t_1 = 3t_2$(or)$t_2 = t_1/3$

$S_1 + S_2 = \frac{t_1^2}{2} + t_1 \times \frac{t_1}{3} - \frac{3}{2} \times \frac{t_1^2}{9}$

$= \frac{2}{3}t_1^2$

$1215 = \frac{2}{3}t_1^2$

$t_1 = 42.69$ s

Total time $t_1 + t_2 = t_1 + \frac{t_1}{3}$

$= \frac{4}{3}t_1 = \frac{4}{3} \times 42.69$

$= 56.92$ s

19. Here $S_1 = \frac{1}{2}a_1 t_1^2$ and $S_2 = \frac{1}{2}a_2 t_2^2$

$V = a_1 t_1$ and $V = a_2 t_2$

$\frac{S_1}{S_2} = \frac{a_1 t_1^2}{a_2 t_2^2} = \frac{vt_1}{vt_2}$

$\frac{S_1}{S_2} = \frac{a_1}{a_2} = \frac{t_1}{t_2}$

20. $x = \frac{1}{2}a \times 10^2 = 50a$

and $x + y = \frac{1}{2} \times a \times 20^2 = 200a$

$y = 200a - x = 200a - 50a = 150a$

$y = 3x$

21. $v^2 - u^2 = 2as$

$S = \frac{6400 - 3600}{2a} = \frac{1400}{a}$

$\frac{S}{2} = \frac{700}{a}$

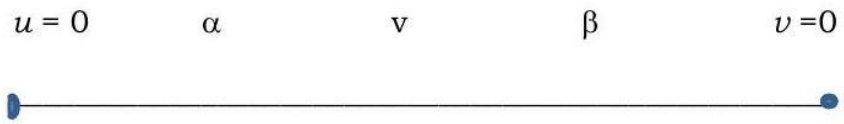
$V^2 - u^2 = 2as$

$V^2 - 60^2 = 2a \times \frac{700}{a}$

$V^2 = 1400 + 3600 = 5000$

$V = \sqrt{5000} = 70.7$ km/hr

22.



A to C

$V = 0 + \alpha t_1$

$V^2 = 0 + 2\alpha S_1$

C to B$0 = V - \beta t_2$ $0 = V^2 - 2\beta S_2$

from (1) $t_1 = \frac{v}{\alpha}$ from (2) $t_2 = \frac{v}{\beta}$

$t_1 + t_2 = t$

$\frac{v}{\alpha} + \frac{v}{\beta} = t$

$V = \frac{\alpha\beta t}{\alpha + \beta}$

from (3) $S_1 = \frac{v^2}{2d}$ from (4) $S_2 = \frac{v^2}{2\beta}$

Total $S_1 + S_2 = S$

$S = \frac{-v^2}{2}\left[\frac{1}{\alpha} + \frac{1}{\beta}\right]$

$S = \frac{\alpha\beta t^2}{2(\alpha + \beta)}$

23. A to B

$S = ut + \frac{1}{2}at^2$

$d = \frac{1}{2}at_1^2$

$t_1 = \sqrt{\frac{2d}{a}}$

A to C $d + d = \frac{1}{2}a(t_1 + t_2)^2$

$t_1 + t_2 = \sqrt{\frac{4d}{a}}$

$t_2 = \sqrt{\frac{4d}{a}} - \sqrt{\frac{2d}{a}}$

A to D $d + d = \frac{1}{2}a(t_1 + t_2 + t_2)^2$

$t_1 + t_2 + t_3 = \sqrt{\frac{6d}{a}}$

$t_3 = \sqrt{\frac{6d}{a}} - \sqrt{\frac{4d}{a}}$

$t_1 : t_2 : t_3 = \sqrt{\frac{2d}{a}} : \left(\sqrt{\frac{4d}{a}} - \sqrt{\frac{2d}{a}}\right) : \left(\sqrt{\frac{6d}{a} - \sqrt{\frac{4d}{a}}}\right) \ldots\ldots\ldots$

$= 1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2}) \ldots\ldots$

24. Train u = 0; a = 2 m/s^2

man 9 m behind, v = ?

displacement of man = diplacement of train

$vt = \frac{1}{2}at^2 + 9$

$v \times 7.5 = \frac{1}{2} \times 2 \times 7.5^2 + 9$

$V\frac{15}{2} = 56.25 + 9$

$\frac{15}{2}v = 65.25$

$V = \frac{65.25 \times 2}{15}$

$V = 8.7$ m/s

25. Let ' t ' be the time after which he catches the bus and let v be his minimum velocity. Here distance covered by both is the same. hence

$Vt = d + \frac{1}{2}at^2 \rightarrow at^2 - 2vt + 2d = 0$

the rots of quadratic equation $t = \frac{v \pm \sqrt{v^2 - 2ad}}{a}$

t is real when $v^2 - 2ad = 0$ for positive and real $V = \sqrt{2ad}$

Level – III

26) B 27) C 28) A 29) A 30) B 31) C

32) B,C 33) a-q; b-r; c-s,t; d-p

26. $S = ut + \frac{1}{2}at^2$

$S = 2 \times 5 + \frac{1}{2} \times 2 \times 25$

$S = 10 + 25 = 35$ m

27. $S_{n^{th}} = u + a\left(n - \frac{1}{2}\right)$

$S_{n^{th}} = 2 + 2\left(5 - \frac{1}{2}\right)$

$= 2 + 2\frac{9}{2}$

$S_{n^{th}} = 11m$

28. $v = u + at$

$= 2 + (2 \times 5)$

$= 2 + 10$

$v = 12$ m/s

29. $S = 2n + 5n2$

$S = ut + \frac{1}{2}at^2$

By companing u = 2 m/s and

$\frac{1}{2}a = 5$

$a = 10$ m/s^2

displacement in 5sec $S = ut + \frac{1}{2}at^2$

$S = 2 \times 5 + \frac{1}{2} \times 10 \times 25$

$S = 10 + 125 = 135$ m

30. $S_{5^{th}} = u + a\left(n - \frac{1}{2}\right)$

$S_{5^{th}} = 2 + 10\left(5 - \frac{1}{2}\right)$

$= 2 + 10 \times \frac{9}{2}$

$= 2 + 45 = 47$ m

31. $v = u + at$

$v = 2 + (10 \times 2)$

$v = 22$ m/s

32. $u = 12$ m/s $a = -2$ m/s^2

$v = u + at$

$o = 12 + (-2)t$

$2t = 12$

$t = \frac{12}{2} = 6$sec

Displacement in 3sec$S = ut + \frac{1}{2}at^2$

$S_3 = (12 \times 3) + \frac{1}{2} \times (-2)$

$S_3 = 36 - 9 = 27m$

Displacement in 6sec

$S_6 = (12 \times 6) + \frac{1}{2}(-2)$

$S_6 = 72 - 36 = 36m$

Displacement between 3sec to 6sec is $S = S_6 - S_3$

$S = 36 - 27 = 9$ m

33. a-q; b-r; c-s,t; d-p

WORKSHEET-8

Level – I

1) A	2) A	3) B	4) B	5) C	6) A	7) A
8) B	9) C	10) B				

Level – II

1) D	2) B	3) A	4) A	5) C	6) A	7) A
8) B	9) D	10) C	11) B	12) B	13) C	14) C 15)
B	16) B	17) D	18) B	19) D	20) C	

1 to 3 : Conceptual

4. $t = 10$sec

$g = 10$ m/s^2

$h = \frac{1}{2}gt^2$

$= \frac{1}{2} \times 10 \times (10)^2$

$= \frac{1}{2} \times 10 \times 100 = 500$ m

5 to 8: Conceptual

9. h = 100 m

g = 10 m/s^2

$t = \sqrt{\frac{2h}{g}}$

$= \sqrt{\frac{2 \times 100}{10}}$

$= \sqrt{20}$

$= 2\sqrt{5}$sec

10. t = 10sec

g = 10 m/s^2

v = gt

$= 10 \times 10$

$= 100$ m/s

11. h = x

t = 10sec

$h = \frac{1}{2} gt^2$

$= \frac{1}{2} \times 10 \times (10)^2$

$= \frac{1}{2} \times 10 \times 100$

$= 500$ m

$\therefore x = 500$ m

v = gt

$= 10 \times 10$

$= 100$ m/s

12. g = 9.8 m/s^2

t = 2sec

u = 0

u = gt

$= 9.8 \times 2 = 2 \times 9.8$ m/s

13. $h = 20$ m

$s = 10$ m

$g = 10$ m/s^2

$s = \frac{1}{2}gt^2$

$10 = \frac{1}{2} \times 10 \times t^2$

$t^2 = \frac{10 \times 2}{10}$

$t = \sqrt{2}$sec

$v = gt$

$= 10 \times \sqrt{2} = 10\sqrt{2}$ m/s

14. $h = \frac{1}{2}gt^2$

$h = \frac{1}{2}g\left(\frac{5t}{6}\right)^2$

$= \frac{1}{2}g\left(\frac{5}{6}\right)^2 t^2$

$= \left(\frac{5}{6}\right)h$

$= \left(\frac{5}{6}\right)^2$ h

15. $h_A = h$

$v_A = u$

$v_A = \sqrt{2gh_A}$

$V = \sqrt{2gh}$

$V_B - V_A = \sqrt{2}\,V - V$

$= (\sqrt{2} - 1)V$

16. $t \propto \sqrt{h}$

$\therefore t_1 : t_2 = \sqrt{h_1} : \sqrt{h_2}$

17. $t_0 = \sqrt{\frac{2h}{g}}$

$h = \frac{h}{2}$

$$t = \sqrt{\frac{2 \times \frac{h}{2}}{g}}$$

$$= \sqrt{\frac{1}{2} \times \frac{2\ h}{g}}$$

$$= \frac{1}{\sqrt{2}} \sqrt{\frac{2\ h}{g}} = \frac{t_0}{\sqrt{2}}$$

$h_B = 2\ h$

$v_B = \sqrt{2gh_B}$

$= \sqrt{2\ g(2\ h)}$

$= \sqrt{2.2gh}$

$= \sqrt{2}\sqrt{2gh}$

$= \sqrt{2}t_0$

18. $S = ut - \frac{1}{2}at^2$

$$6 = v \times 0.2 - \frac{1}{2} \times 10 \times (0.2)^2$$

$$6 = 0.2v - \frac{1}{2} \times 10 \times 0.04$$

$$6 = 0.2v - \frac{1}{2} \times 10 \times 0.02$$

$6 = 0.2\ V - 0.2$

$6 + 0.2 = 0.2v$

$$V = \frac{6.2}{0.2} = 31\ \text{m/sw}$$

$$h = \frac{V^2}{2\ g} = \frac{(31)^2}{2 \times 10}$$

$= 96 + 480.5 = 48.05$ m

19. $h = \frac{1}{2}gt^2$

$$0.64h = \frac{1}{2}g(t-1)^2$$

$0.64 \times \frac{1}{2}gt^2 = \frac{1}{2}\ g(t-1)^2\ \ 0.64t^2 = (t-1)^2$

$0.64t^2 = t^2 + 1 - 2t$

$t^2 - 0.64t^2 - 2t + 1 = 0$

$0.36t^2 - 2t + 1 = 0$

$36t^2 - 200t + 100 = 0$

$9t^2 - 50t + 25 = 0$

$9t^2 - 5t - 45t + 25 = 0$

$t(9t - 5) - 5(9t - 5) = 0$

$\therefore t - 5(9t - 5) = 0$

$\therefore t = 5 \text{ or } \frac{5}{9}$

$t = 5$

$h = \frac{1}{2} g t^2$

$= \frac{1}{2} \propto (5)^2$

$= \frac{1}{2} \propto w \propto 25$

$= 125 \text{ m}$

20. $t = 5\text{sec}$

$h = \frac{1}{2} g t^2$

$= \frac{1}{2} \times 10 \times (5)^2$

$= \frac{1}{2} \times 10 \times 25$

$= 125 \text{ m}$

$t_1 = 3\text{sec}$

$h_1 = \frac{1}{2} \text{ g} t_1^2$

$= \frac{1}{2} \times 10 \times (3)^2$

$= \frac{1}{2} \times 10 \times 9$

$= 45 \text{ m}$

$h_2 = h - h_1$

$= 125 - 45$

$= 80 \text{ m}$

$t_2 = \sqrt{\frac{2gh_2}{g}}$

$= \sqrt{\frac{2 \times 80}{10}}$

$\sqrt{16} = 4\text{sec}$

Level – III

21) A,B,C,D	22) C	23) A	24) A	25) a-s; b-r; c-q; d-p
26) 60m	27) A,C	28) A,B,C,D		29) a-q; b-p; c-r; d-s,t

21-24: Conceptual

25. a-s; b-r; c-q; d-p

26. $t_1 = 2 + 2 = 4\text{sec}$

$h_1 = \frac{1}{2}gt_1^2$

$= \frac{1}{2} \times 10 \times (4)^2$

$= \frac{1}{2} \times 10 \times 16 = 80\ \text{m}$

$h_1 - h_2 = 80 - 20 = 60\ \text{m}$

$t_2 = 2\text{sec}$

$h_2 = \frac{1}{2}gt_2^2$

$= \frac{1}{2} \times 10 \times (2)^2 = 20\ \text{m}$

27. $h = 20\ \text{m}\ t = \sqrt{\frac{2h}{g}}$

$= \sqrt{\frac{2 \times 20}{10}}$

$= \sqrt{4}$

$= 2\text{sec}$

$V = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 20}$

$= \sqrt{(20)^2}$

$= 20\ \text{m/s}$

28. $h = 50\ \text{m}$

$g = 10\ \text{m/s}^2$

$t = \sqrt{\frac{2h}{g}}$

$= \sqrt{\frac{2 \times 50}{10}}$

$= \sqrt{10}\text{sec}$

$V = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 50}$

$\sqrt{1000}$ m/s

$h^1 = \frac{h}{2} = \frac{50}{2} = 25$ m

$t^1 = \sqrt{\frac{2\ h^1}{g}}$

$= \sqrt{\frac{2 \times 25}{10}} = \sqrt{\frac{50}{10}} = \sqrt{5}\text{sec}$

$V^1 = \sqrt{2gh^1}$

$= \sqrt{2 \times 10 \times 25}$

$= \sqrt{500}$ m/s

29. a-q; b-p; c-r; d-s,t

WORKSHEET-9

Level – I

1) A	2) B	3) A	4) B	5) A	6) A	7) B
8) B	9) A	10) B	11) A	12) A		

Level – II

1) A	2) B	3) A	4) D	5) C	6) A	7) D
8) C	9) C	10) B	11) C	12) A	13) D	14) B 15)
C	16) B	17) D	18) C	19) D	20) B	

1. $H_{max} = \frac{-u^2}{29} = \frac{10^2}{2(10)} = 5$ m

2. $H_{more} = \frac{u^2}{2\ g} \Rightarrow u^2 = H \cdot 2\ g = 5(2)(10) = 100$

   $\Rightarrow u = \sqrt{100} = 10$ m/s

3. $T = \frac{2u}{g} = \frac{2(9.8)}{9.8} = 2\text{sec}$

4. $H = \frac{u^2}{2\,g} = \frac{(20)^2}{9(10)} = 20$ m

5. $H = \frac{u^2}{2\,g} \Rightarrow 4 = \sqrt{2gH} = \sqrt{2(10)(80)}$

$= \sqrt{1600} = 40$ m/s

6. $H_{max} = \frac{u^2}{2\,g} = \frac{(15)^2}{2(10)} = \frac{225}{20} = 11.25$ m

7. $t_a = \frac{u}{g} = \frac{10}{10} = 1$sec

8. $t_a = \frac{u}{g} \Rightarrow u = t_a \times g = 3 \times 9.829.4$ m/s

9. $t_a = \frac{u}{g} = \frac{19.6}{9.8} = 2$sec

10. $t_d = \frac{u}{g} = \frac{19.6}{9.8} = 2$sec

11. $H_{max} = \frac{u^2}{2\,g} \Rightarrow u = \sqrt{2\,gH_{max}} = \sqrt{2(9.8)(90)}$

$= \sqrt{196 \times 9} = 14 \times 3 = 42$ m/s

12. $T = \frac{2u}{g} = \frac{2(14)}{10} = 2.8$sec

13. $T = \frac{2u}{g} = \frac{2(40)}{10} = 8$sec

14. $t_a = \frac{u}{g} = \frac{39.2}{9.8} = 4$sec

15. $t_a = \frac{u}{g} \Rightarrow u = t_a \times g = 5 \times 9.8 = 49$ m/s

16. $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 19.6} = 19.6$ m/s

17. Body reaches its initial position after some time so, its displacement will be zero due to this the average velocity will become zero.

18. $H = \frac{u^2}{2\,g}\frac{(40)^2}{2(10)} = \frac{1600}{20} = 80$ m

So, total distance = 80 + 80

[up and down]

= 160 m

19. Given T = 6sec

$\Rightarrow \frac{2u}{g} = 6 \Rightarrow u = \frac{6 \times 10}{2} = 30$ m/s

So, $h = ut - \frac{1}{2}gt^2$

$\Rightarrow h = 30(4) - \frac{1}{2}(10)(4)^2$

$\Rightarrow h = 120 - 80 = 40$

$\therefore h = 40$ m

20. Height of tower = 60 m

initial velocity = 20 m/s

final velocity = 0

using $v = u - gt$

$\Rightarrow 0 = 20 - 10t \Rightarrow 10t = 20 \Rightarrow t = 2sec$

Now, time of ascent = time of descent therefore, the ball pass the man after '4' seconds going downward.

Level – III

21) A,B,C,D 22) A 23) B 24) A 25) A

26) a-q; b-p; c-t; d-s 27) 5 28) 20 29) A,B

4. TYPES OF FORCE

WORKSHEET-1

Level – I

1) A	2) D	3) B	4) A	5) D	6)A	7) D
8) C	9)A	10) A				

Level – II

1)A	2)B	3)A	4) C	5) D	6) C	7) B
8) D	9) D	10) D	11) B	12) C	13) D	14) B
15) A	16) C	17) B				

1-13: Conceptual

14. $F = \frac{Gm_1\, m_2}{r^2}$

15. $F = \frac{Gm_1\, m_2}{r^2}$

$$F = \frac{6.67 \times 10^{-11} \times 1\ \text{kg} \times 2\ \text{kg}}{0.5 \times 0.5}$$

$$F = 5.3 \times 10^{-10}\ \text{N}$$

16. We know that $g^1 = \frac{GM}{(R+h)^2}$

At $h = 36000\ \text{km} \cdot g^1 = \frac{GM}{(36000+6400)^2}$

$$\therefore \frac{g^1}{g} = \frac{6400 \times 6400}{42400 \times 42400}$$

$$= \frac{256}{106 \times 106} = 0.0227$$

$$\Rightarrow g^1 = 0.0227 \times 9.8 = 0.223$$

[taking $g = 9.8\ \text{m/s}^2$] at the surface of earth]

A 120 kg equipment placed in a geos tationary satellite will have weight

$Mg' = 0.233 \times 120 = 26.79 = 27\ \text{N}$

Level – III

18) A,B,C	19) a-p,q,s; b-r; c-p,q,s; d-p,q,s		20) 5	21) B	
22) B	23) D	24) B,C,D	25) B	26) B	27) C
28) 8	29) a-r; b-p; c-q; d-s		30) B		

27. $m_1 = 2\ \text{kg},\ m_2 = 4\ \text{kg}, r = 2\ \text{m}$

$$F = \frac{Gm_1\, m_2}{r^2}$$

$$F = \frac{6.67 \times 10^{-11} \times 2 \times 4}{2}$$

$F = 2G$

28. $F = \frac{Gm_1\ m_2}{r^2}$

$$F = \frac{6.67 \times 10^{-11} \times 5 \times 6}{5 \times 5}$$

$F = 8.004 \times 10^{-11}$ N

29. a-r; b-p; c-q; d-s

30. $F = \frac{Gm_1m_2}{r^2}$

$r = 1/2$

$F = \frac{Gm_1m_2}{(1/2)^2}$

$F = \frac{Gm_1m_2}{1/4}$

$F' = 4F$

WORKSHEET-2

Level – I

1) B	2) A	3) C	4) D	5) D	6) B	7) C
8) A	9) B					

Level – II

1) B	2) B	3) B	4) B	5) A	6) C	7) C
8) A	9) D	10) C	11) A	12) C	13) C	14) D
15)A	16) A	17) B	18) C	19) A	20) A	21) C

Level – III

22) 1,2,4	23) a-q,r,s; b-p; c-t; d-p	24) 1	25) 3	26) 1	27) 3
28) 1	29) a-r,s; b-r,s; c-r,s; d-r,s				

1-30: Conceptual

WORKSHEET-3

Level – I

1) B	2) A	3) D	4) C	5) C	6) D	7) A
8) A	9) C	10) C				

Level – II

1) C	2) A	3) B	4) B	5) B	6) C	7) D
8) B	9) A	10) C	11) A	12) B	13) B	14) D
15) A	16) A	17) B	18) A	19) B	20) C	

Level – III

21) A,B,C	22) a → $p, b \rightarrow q, c \rightarrow r, d \rightarrow s$	23) C	24) C	25) D
26) a → q, bd → r, c → s, d → p	27) A,C,D	28) B	29) A	30) B

1-27: Conceptual

28. w_{cube} = volume of cube × density of cube × g

$= 1000\ cm^3 \times 0.91 \times g$

$= 910\ g \times g$

$w_{cube} = 0.91 \times 10 = 9.1\ N$

29. $f_b = v\rho_{liquid}\ g$

$w_{cube} = v\rho_{water}\ g = 9.1 \times 10^{-4}\ m^3$

30. $v\rho_{water}\ g = 9.1 \times 10^{-3}\ N$

5. NEWTON'S LAW MOTION

WORKSHEET-1

Level – I

1) C	2) D	3) B	4) C	5) C	6) A	7) B
8) B	9) A	10) B				

Level – II

1) C	2) C	3) A	4) B	5) C	6) D	7) B
8) B	9) C	10) B	11) A	12) C	13) D	14) A
15) D	16) D					

Level – III

17) A,B,C	18) A	19) A	20) B	21) D	22) a-q,r; b-p; c-s; d-t
23) A	24) A	25) A			

WORKSHEET-2

Level – I

1) D	2) D	3) B	4) A	5) A	6) C	7) D

Level – II

1) A	2) D	3) D	4) C	5) D	6) D	7) D
8) C	9) A	10) D	11) B	12) D	13) B	14) D
15) B	16) B					

1. $P = mv$

$v_1 = 2v$

$m_1 = \frac{m}{2}$

$P_1 = m_1 v_1 = \frac{m}{2} \times 2v$

$P_1 = mv$

$P_1 = P$

2. $P_1 = \sqrt{2k_1\ m_1}$

$P_2 = \sqrt{2k_2\ m_2}$ $P_1 = P_2$ (Given)

$\sqrt{2k_1\ m_1} = \sqrt{2k_2\ m_2}$

$2k_1\ m_1 = 2k_2\ m_2$

$\frac{k_1}{k_2} = \frac{m_2}{m_1}$

$\because k_1 : k_2 = m_2 : m_1$

3. $k = p(\text{ given })$

$\frac{1}{2}mv^2 = mv$

$v = 2\ m/s$

4. $P = \sqrt{2\ km}$

$k_1 = 2k$

$P_1 = \sqrt{2k_1\ m}$

$P_2 = \sqrt{2(2k)m} = \sqrt{2} \cdot \sqrt{2\ km}$

$\because P_2 = \sqrt{2}P$

5. $m = 1\ kg$

$P = 2\ N - S$

$K.E = \frac{P^2}{2\ m} = \frac{2(2)}{2 \times 1}$

$\therefore K.E = 2\ J$

6. $m_1 = 2\ m$

$m_2 = m$

$\frac{k_1}{k_2} = \frac{8}{1}$

$K_1 = \frac{P_1^2}{2\ m_1}; K_2 \frac{P_2^2}{2\ m_2}$

$\frac{K \cdot E_1}{K \cdot E_2} = \frac{p_1^2}{2\ m_1} \times \frac{2\ m_2}{P_2^2}$

$\frac{8}{1} = \frac{P_1^2}{P_2^2} \times \frac{m}{2\ m}$

$= \frac{P_1^2}{P_2^2} = \frac{16}{1}$

$= \frac{P_1}{P_2} = \frac{4}{1}$

$\therefore P_1 : P_2 = 4 : 1$

7. $\frac{K_1}{K_2} = \frac{4}{1}$

$P_1 = P_2$ (Given)

$\sqrt{2k_1\ m_1} = \sqrt{2k_2\ m_2}$

$2k_1\ m_1 = 2k_2\ m_2$

$\frac{k_1}{k_2} = \frac{m_2}{m_1}$

$\frac{4}{1} = \frac{m_2}{m_1}$

$\frac{m_1}{m_2} = \frac{1}{4}$ or $1:4$

11. $\frac{K\cdot E_1}{K\cdot E_2}=\frac{m_1}{m_2}\times\frac{v_1^2}{v_2^2}$

$\Rightarrow \frac{K\cdot E_1}{K\cdot E_2}\times\frac{m_2}{m_1}=\frac{v_1^2}{v_2^2}$

$\Rightarrow \frac{1}{3}\times\frac{1}{3}=\frac{v_1^2}{v_2^2}$

$\Rightarrow v_1 : v_2 = 1:3$

12. $K\cdot E=\frac{p^2}{2\,m}$

$K.E=\frac{20000\times 20000}{2\times m}$

$\Rightarrow 200=\frac{20000\times 20000}{2\times m}$

$\Rightarrow m=\frac{20000\times 20000}{2\times 200}$

$\Rightarrow m=\frac{100\times 10000}{1}$

$\Rightarrow m = 10^6\ g$

$\Rightarrow m = 1000\ kg$

13. $K\cdot E=\frac{P^2}{2\,m}$ $K.E=(P^2)^2=P^4$

$\Rightarrow 4$ times

14. $K.E=\frac{P^2}{2\,m}$

$\Rightarrow \frac{m_1}{m_2}=\frac{P_1^2}{P_2^2}\times\frac{K\cdot E_2}{K\cdot E_1}$

$\Rightarrow \frac{m_1}{m_2}=\frac{1}{4}\times\frac{1}{2}$

$\Rightarrow \frac{m_1}{m_2}=\frac{1}{8}$ or $1:8$

15. $K.E=\frac{1}{2}mv^2$

$\Rightarrow \frac{K\cdot E_1}{K\cdot E_2}=\frac{V_1^2}{V_2^2}$

$\Rightarrow \frac{K\cdot E_1}{K\cdot E_2}=\frac{V^2}{4\,V^2}$

$\Rightarrow \frac{K\cdot E_1}{K\cdot E_2}=\frac{1}{4}$ or $1:4$

16. $T\cdot E=\frac{1}{2}mv^2=\frac{1}{2}\times 3\times 100=150$ J

Lost $= 15$ J

left over $= 150-15=135$ J

$mgh = E$

$\Rightarrow 3\times 10\times h = 135$

$\Rightarrow h = \frac{135}{3 \times 10} = \frac{27}{6}$

$\therefore h = \frac{9}{2} = 4.5$ m

Level – III

17) D 18) A,B 19) a-p; b-q; c-r; d-s,t 20) B 21) D

22) D 23) A,C 24) A 25) 3

23. $F \times t = I = m(v - u)$

$= 150 \times 10^{-3}(0 - 20) = 3.0\text{NS}$

$\Rightarrow F = \frac{3}{0.1} = 30N$

24. Volume of water striking the wall per sec = av

V is the velocity of water

But in this problem as water making angle

Volume of water striking per sec = $av\cos\theta$

Mass striking/sec

$= av\cos\theta \times \rho = av\rho\cos\theta$

As bouncing back, so momentum change/sec

= force = 2mv = $av\rho\cos\theta \times v$

$= 2av^2\rho\cos\theta$

25. $m = 600$ g = 0.6 kg

Given that ball bounces with same velocity and with same angle.

$\therefore u\sin\theta = v\sin\theta$

Change in momentum

$= mu\sin\theta - (-mv\sin\theta)$

$= mu\sin\theta + mv\sin\theta$

$= 2mu\sin\theta$ ($\because u = v$)

$\Delta P = 2 \times 0.6 \times 5 \times \frac{1}{2} = 3\text{kgms}^{-1}$

WORKSHEET-3

Level – I

1) B 2) A 3) C 4) C 5) C 6) A 7) A

Level – II

1) B 2) C 3) A 4) C 5) A 6) C 7) B

8) B 9) C 10) A 11) B 12) D 13) B

14) A,B,C,D 15) B 16) C 17) D 18) a-q; b-p; c-t; d-r,s

19) 105 20) D 21) D 22) D 23) C 24) A 25) B

1. $m = 200\ g = 0.2\ kg$

   $u = 15\ m/s$

   $v = 25\ m/s$

   $t = 2.5sec$

   $f = ma$

   $f = m\left[\frac{v-u}{t}\right]$

   $f = 0.2\left[\frac{25-15}{2.5}\right]$

   $f = \frac{0.2 \times 10}{2.5} = \frac{20}{25}$

   $\Rightarrow 0.8\ N$

2. $F = 0.6\ N$

   $u = 5\ m/s$

   $v = 6\ m/s$

   $t = 2sec$

   $f = ma = m\left[\frac{v-u}{t}\right]$

   $0.6 = m\left[\frac{6-5}{2}\right]$

   $m = 2 \times 0.6$

   $\Rightarrow 1.2\ kg$

3. $f = 12\ N$

   $t = 4\ S$

   $f = m\left(\frac{v-u}{t}\right)$

   $12 = \frac{mv-mu}{4}$

   $48 = m(v-u)$

   $48\ kg\frac{m}{s} = \Delta p$

4. $a = \frac{5\ cm}{s^2}$

   $m = 20\ g$

   $f = ma = 20 \times 5$

   $f = 100\ g\frac{cm}{s^2} = 100\ dynes$

   $= 100 \times 10^{-5}\ N = 10^{-3}\ N$

5. $f_1 = ?$

   $m_1 = 2\ kg$

   $a_1 = 2.5\ m/s^2$

   $f_2 = ?$

$m_2 = 5$ kg

$a_2 = 2$ m/s^2

$f_1 = m_1 a_1 = 2 \times 2.5 \Rightarrow 5$ N

$f_2 = 5 \times 2 \Rightarrow 10$ N

6. For a body A:

$f_1 = m_1 a_1$

$f_2 = 2\ m_1$

For a body B :

$f_2 = m_2 a_2 = m_2 \times 5 = 5\ m_2$

$f_1 = f_2 =$ (Given)

$2\ m_1 = 5\ m_2$

$\frac{m_1}{m_2} = \frac{5}{2}$

7. $m = 5$ g $= 5 \times 10^{-3}$ kg

$u = 0$

$v = 120$ m/s

$s = 2$ m

$v^2 - u^2 = 2as$ $(120)^2 - (0)^2 = 2 \times \mathbf{a} \times 2$

$a = \frac{14400}{4} = 3600$ m/s^2

$f = ma$

$f = 5 \times 10^{-3} \times 3600 \Rightarrow 18$ N

8. $v^2 - u^2 = 2as$

$0 - u^2 = 2as$

$s = \frac{u^2}{2a}$

$\frac{u^2}{2a} =$ constant

$\left(\frac{u_1}{u_2}\right)^2 = \frac{a_1}{a_2} \rightarrow 1$

$f = ma$

$f \alpha a$

$\Rightarrow \frac{f_1}{f_2} = \frac{a_1}{a_2} \longrightarrow^2$

$\frac{f_1}{f_2} = \frac{u_1^2}{u_2^2}$

$f_1 = 100$ N, $u_1 = 20$ m/s, $u_2 = 30$ m/s

$\frac{100}{f_2} = \frac{(20)^2}{(30)^2}$

$$\frac{100}{f_2} = \frac{400}{900}$$

$$f_2 = \frac{100 \times 900}{400} \Rightarrow 225 \text{ N}$$

9. $m = 5$ kg

$f = mg = 5 \times 9..8 = 5$kgwt

$\because$ 1kgwt $= 9.8$ N

10. $f = ma$

$$f = m\left[\frac{v - u}{t}\right]$$

$f = 100N; v = 0; t = 0.25$sec

$$100 = \frac{m(0 - u)}{0.25}$$

$$mu = 25 \text{ kg}\frac{m}{s}$$

11. $m = 5$ kg

$a = 4$ m/s^2

$t = 5$sec

$v = u + at$

$v = u + 4 \times 5 = 20$ m/s

momentum after ' 5 ' sec is

$$p = mv = 5 \times 20 \Rightarrow 100 \text{ kg}\frac{m}{s}$$

12. $m = 150$ g $= 150 \times 10^{-3}$ kg

$n = 20$

$v = 800$ m/s

$t = 1$sec

$$f = \frac{nma}{t} = \frac{nmv}{t^2}$$

$f = 20 \times 150 \times 10^{-3} \times 800$

$f = 20 \times 15 \times 8$

$f = 2400$ N

15. $a = \frac{v-u}{t}$

$$a = \frac{6 - 5}{0.1} = \frac{1}{0.1} = \frac{1}{\frac{1}{10}} = 10 \text{ m/s}^2$$

16. $m(v - u) = \Delta p$

$2(6 - 5) = \Delta p$

$2\ \text{kg}\frac{\text{m}}{\text{s}} = \Delta\text{p}$

17. $F = \frac{m(v-u)}{t}$

$F = \frac{m(v-u)}{t} = \frac{2}{0.1} = 20\ \text{N}$

20. $36\text{kmph} \rightarrow 36 \times \frac{5}{18} = 10\ \text{m/s} = \text{u}$

$18\text{kmph} \rightarrow 18 \times \frac{5}{18} = 5\ \text{m/s} = \text{v}$

$\text{a} = \frac{5-10}{0.5} = \frac{5}{5} = \frac{50}{5} = 10\ \text{m/s}^2$

$\text{f} =, \text{a} = 8 \times 10 \Rightarrow 80\ \text{N}$

21. $S_1 = u(1) + \frac{1}{2}a(1^2) = 3$

$\Rightarrow 2u + a = 6$

$S_2 = S_1 + 1 = u(2) + \frac{1}{2}a(2^2) = 3 + 1 = 4$

$2u + 2a = 4$

$\therefore a = -2m/s^2; u = 4m/s$

$F = ma = 2(-2) = -4N$

22. $\text{f} = \text{ma}$

$\text{m} = \frac{\text{f}}{\text{a}}$

Given $\text{f} = {}'\text{g}\ '\ \text{N}$

so, $\text{m} = \frac{\text{g}}{\text{a}}$

$\text{w} = \text{mg}$

$\text{w} = \frac{\text{g}}{\text{a}} \times \text{g} = \frac{\text{g}^2}{\text{a}}$

23. $\text{F} = 48\ \text{N}$

$\text{m} = 16\ \text{kg}$

$\text{F} = \text{ma}$

$\text{a} = \frac{\text{F}}{\text{m}} = \frac{48}{16} = 3\ \text{m/s}^2$

24. $\text{S} = \frac{1}{2}\text{att}^2$

$= \frac{1}{2} \times 3 \times (2.50)^2$

$= \frac{1}{2} \times 3 \times 6.25$

$= 9.38\ \text{m}$

25. $v^2 - u^2 = 2aS$

$\text{v}^2 = 2 \times 3 \times 9.38$

$v^2 = 56.28$

$v = \sqrt{56.28} = 7.5$ m/s

WORKSHEET-4

Level – I

1) B	2) B	3) D	4) C	5) C	6) D	7) D
8) A	9) B	10) A				

Level – II

1) A	2) A	3) B	4) A	5) D	6) A	7) C
8) B	9) B	10) B	11) A	12) C	13) C	14) D
15) D	16) C	17) D				

9. $m_1 = 3$ kg $u_1 = 4$ m/s

$m_2 = 4$ kg $u_2 = 2$ m/s

Initial momentum $= m_1u_1 - m_2u_2$

$= 3(4) - 4(2)$

$= 12 - 8 = 4$kgm/s

10. $m_1 = 100$ kg $u = 30$ m/s

$v = 20$ m/s

Change in momentum = Impulse

$= m(v - u)$

$mv(-mu) = 2000 + 3000 = 5000$Ns

11. $m = 2.5$ kg, $u = 1$

Impulse = mass × velocity

$= 2.5(1) = 2.5$NS

12. $m_1 = a$ $u_1 = b$

$m_2 = c$ $u_2 = 0$

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} = \frac{a(b) + 0}{a + c}$$

$$v = \frac{ab}{a + c}$$

13. $m_1 = 2$ kg $u_1 = 40$ m/s $u_2 = 0$

$v = 25$ m/s $m_2 = ?$

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$2(40) + 0 = (2 + m_2)25$

$80 = 50 + 25\ m_2$

$80 - 50 = 25\ m_2$

$$\frac{30}{25} = m_2$$

$m_2 = \frac{6}{5} = 1.2$ kg

$m_2 = 1.2$ kg

14. $m_1 = m\ u_1 = v$

$m_2 = 2\ m\ u_2 = 2v$

$m_3 = 3\ m\ u_3 = 3v$

$m_4 = 4\ m\ u_4 = 4v$

$m_5 = 5\ m\ u_5 = 5v$

$$v = \frac{m_1u_1 + m_2u_2 + m_3u_3 + m_4u_4 + m_5u_5}{m_1 + m_2 + m_3 + m_4 + m_5}$$

$$v = \frac{m(v) + 2\ m(2v) + 3\ m(3v) + 4\ m(4v) + 5\ m(5v)}{m + 2\ m + 3\ m + 4\ m + 5\ m}$$

$$= \frac{mv + 4mv + 9mv + 16mv + 25mv}{15\ m}$$

$$= \frac{55mv}{15\ m} = \frac{11v}{3}$$

$$v = \frac{11v}{3}\ m/s$$

15. $m_1 = 10000$ kg $u_1 = 2$ m/s

$m_2 = 20{,}000$ kg $u_2 = 0$

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} = \frac{10{,}000(2) + 0}{10000 + 20000} = \frac{20000}{30000}$$

$$v = \frac{2}{3}$$

$v = 0.66$ m/s

16. $m_1 = 5 \times 200$gm $u_1 = 10$ m/s

$$= \frac{1000}{1000} = 1 \text{ kg } u_2 = 0$$

$m_2 = 3$ kg

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

$$v = \frac{1(10) + 3(0)}{1 + 3} = \frac{10}{4}$$

$v = 2.5$ m/s

17. $m_1 = 10^5$ kg $u_1 = 0.5$ m/s

$m_2 = 2 \times 10^5$ kg $u_2 = 0$

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} = \frac{10^5(0.5) + 0}{10^5 + 2 \times 10^5}$$

$$= \frac{10^5(5/10)}{10(1 + 2)} = \frac{5}{10^{(3)}}$$

$$= \frac{5}{30} = \frac{1}{6} = 0.166 \text{ m/s}$$

Level – III

18) C 19) a-p,q; b-p,q; c-s; d-t 20) B 21) A 22) D

23) A 24) A 25) A,B

18. $m_1 = 1.67 \times 10^{-27}$ kg $u_1 = 3 \times 10^6$ m/s

$m_2 = 3.34 \times 10^{-27}$ kg $u_2 = 0$

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

$$= \frac{1.67 \times 10^{-27} \times 3 \times 10^6 + 0}{1.67 \times 10^{-27} + 3.34 \times 10^{-27}}$$

$$= \frac{1.67 \times 10^{-27} \times 3 \times 10^6}{10^{-27}(1.67 + 3.34)}$$

$$= \frac{5.01 \times 10^6}{5.01} = 10^6$$

$v = 10^6$ m/s

20. $m_1 = 500$ kg $u_1 = 4$ m/s

$m_2 = 1500$ kg $u_2 = 2$ m/s

$$v = \frac{m_1u_1 + m_2 + u_2}{m_1 + m_2} = \frac{500(4) + 1500(2)}{500 + 1500}$$

$$= \frac{2000 + 3000}{2000} = \frac{5000}{2000} = \frac{5}{2} = 2.5 \text{ m/s}$$

22. $m_1 = \frac{2 \times 10}{2} = 10^4$ kg

$u_1 = 0.5$ m/s

$u_2 = 0.4$ m/s

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

$$= \frac{2 \times 10^4(0.5) - 10^4(0.4)}{10^4 \times 2 + 1}$$

$$= \frac{10000 - 4000}{30000} = \frac{6000}{30000} = \frac{1}{5}$$

$v = 0.2$ m/s

23. $F = \frac{d}{dt}(mv) = n\left(\frac{dm}{dt}\right)$

$$210 = 300\left(\frac{dm}{dt}\right)$$

$\frac{dm}{dt}$ = rate of combustion $= \frac{210}{300} = 0.7$ kg/s

24. $V = -\frac{mv}{M} = -\frac{20 \times 10^{-3} \times 100}{8} = -0.25$ m/s

25. Force exerted on the wall

$F = Av^2\rho$

$$F = \left(72 \times \frac{5}{18}\right)^2 \times 100 \times 1.2 = 48000N$$

WORKSHEET-5

Level – I

1) A 2) B 3) B 4) B 5) A 6) A 7) B 8) A

Level – II

1) D 2) B 3) B 4) A 5) B 6) B 7) A
8) C 9) B 10) B 11) A 12) C 13) D 14) A
15) A 16) D 17) B 18) A 19) A 20) A 21) A
22) A

1. $m_1 = 5$ kg

$m_2 = 4.8$ kg

$m_1 > m_2$

$g = 9.8\ m/s^2$

$$a = \left[\frac{m_1 - m_2}{m_1 + m_2}\right] \times 9$$

$$= \left[\frac{5 - 4.8}{5 + 4.8}\right] \times 9.8$$

$$a = \frac{0.2}{9.8} \times 9.8$$

2. $m_1 = 2$ kg

$m_2 = 1$ kg

$m_1 > m_2$

$g = 10\ m/s^2$

$$a = \left[\frac{m_1 - m_2}{m_1 + m_2}\right] g$$

$$= \left[\frac{2 - 1}{2 + 1}\right] \times 10$$

$$= \frac{1}{3} \times 10$$

$$a = \frac{10}{3}\ m/s^2$$

$u = 10$ m/s, $a = \frac{10}{3}\ m/s^2$

$$s = ut + \frac{1}{2}at^2$$

$$\frac{10}{3} = 0 + \frac{1}{2} \times \frac{10}{3} \times t^2$$

$$t^2 = 2$$

$t = \sqrt{2}sec$

3. $m_1 = 1\ kg$

$m_2 = 2\ kg$

$$T = \frac{F}{m_1 + m_2} = \frac{F}{1+2}$$

$T = F/3$

4. $m_1 = 4\ kg$

$g = 10\ m/s$

$T = 60\ N$

$T - mg = ma$

$60 - 4 \times 10 = 4a$

$$a = \frac{20}{4}$$

$a = 5\ m/s^2$

5. $m_1 = 2\ kg,\ m_2 = 2\ kg$

$g = 10\ m/s^2$

force acting on pully $= \left[\frac{\sqrt{2}\ m_1\ m_2}{m_1+m_2}\right]$

$$= \frac{\sqrt{2} \times 2 \times 3}{2+3} \times 10 = \frac{6\sqrt{2}}{5} \times 10$$

$F = 12\sqrt{2}\ N$

11. $m_1 = 4\ kg,\ F = 20N$

$m_2 = 2\ kg$

$$a = \frac{f}{m_1 + m_2} = \frac{20}{4+2} = \frac{20}{6}$$

$$a = \frac{10}{3}\ m/s^2$$

$$F_{12} = \frac{m_2\ F}{m_1 + m_2} = \frac{2 \times 20}{4+2} = \frac{40}{6} = \frac{20}{3}\ N$$

$$F_{12} = \frac{20}{3}\ N$$

13. $m_1 = 4\ kg$

$m_2 = 5\ kg$

$$a = \frac{m_2\ g}{m_1 + m_2} = \frac{5\ g}{4+5}$$

$$a = \frac{5}{9}g$$

$$F = (m_1 + m_2)a = 6\left(\frac{5}{9}\right)10 = 33.3N$$

14. $m_1 = 4\ kg$

$m_2 = 2\ kg$

$F = 20\ N$

$$F_{12} = \frac{m_2\ F}{m_1 + m_2}$$

$$= \frac{2 \times 20}{4 + 2} = \frac{40}{6} = \frac{20}{3}$$

$$F_{12} = \frac{20}{3}\ N$$

15. $m_1 = 30\ g,\ m_2 = 40\ g$

$$a = \left[\frac{m_2 - m_1}{m_1 + m_2}\right] \times 9$$

$$a = \left[\frac{40 - 30}{40 + 30}\right] \times 10$$

$$a = \frac{10}{70} \times 10 = 1.4\ m/s^2$$

16. $m_1 = 6\ kg$

$m_2 = 10\ kg$

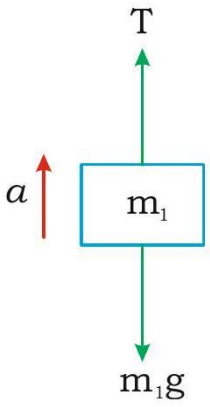
$$T = \frac{2\ m_1\ m_2}{m_1 + m_2} \times 9$$

$$T = \frac{2 \times 6 \times 10}{6 + 10} \times 9.8$$

$$T = \frac{2 \times 6 \times 10}{16} \times 9.8$$

$T = 3 \times 5 \times 4.9 = 73.5\ N$

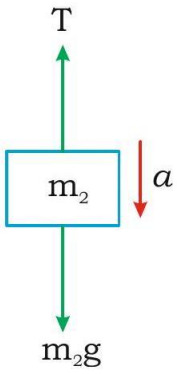
17. Free body diagram for m_1,



$F_{net} = t - m_1\ g$

$m_1 a = T - m_1\ g \rightarrow (1)$

Free body diagram from m_2



$F_{net} = m_2\ g - T$

$m_2a = m_2\ g - T \rightarrow (2)$

adding (1) & (2)

$m_1a = T - m_1\ g$

$m_2a = T - m_2\ g$

$a(m_1 + m_2) = m_2\ g - m_1\ g$

$a = \left[\frac{m_2 - m_1}{m_1 + m_2}\right] g$

Substitute $a = \left[\frac{m_2+m_1}{m_1+m_2}\right] g$ in(1)

$\Rightarrow m_1 \left[\frac{m_2-m_1}{m_1+m_2}\right] g + m_1\ g = T.$

$\Rightarrow m_1\ g\left[\frac{m_2 - m_1}{m_1 + m_2}\right] + m_1\ g = T$

$\Rightarrow m_1\ g\left[\frac{m_2 - m_1}{m_1 + m_2} + 1\right] = T$

$\Rightarrow m_1\ g\left[\frac{m_2 - m_1 + m_1 + m_2}{m_1 + m_2}\right] = T$

$\Rightarrow m_1\ g\left[\frac{2\ m_2}{m_1 + m_2}\right] = T$

$T = \left[\frac{2\ m_1\ m_2}{m_1 + m_2}\right] g$

18. $m_1 = 300\ g = 0.3\ kg$

$m_2 = 600\ g = 0.6\ kg$

$a = \left[\frac{m_2-m_1}{m_1+m_2}\right] \times g\ a = \left[\frac{0.6-0.3}{0.6+0.9}\right] \times 10$

$a = \frac{0.3}{0.9} \times 10$

$a = \frac{30}{9} \times \frac{10}{3}\ m/s^2$

$a = \frac{10}{3}\ m/s^2$

$t = 2sec$

$S = \frac{1}{2} a t_5^2$

$S = \frac{1}{2} \times \frac{10}{5} \times 2^2$

$S = \frac{20}{3}$ m

19. $m_1 = 3$ kg

$m_2 = 2$ kg

$m_3 = 1$ kg

$a = \frac{F}{m_1 + m_2 + m_3}$

$a = \frac{12}{3 + 2 + 1}$

$a = \frac{12}{6} = 2$

$a = 2$ m/s^2

$F_{12} = 6$ N

$F_{23} = m_3 a = 1 \times 2$

$F_{23} = 2$ N

Net force on 2 kg

$F_{net} = F_{12} - F_{23} = 6 - 2$

$F_{net} = 4$ N

20. Given $m_1 = 4$ kg

$m_2 = 2$ kg

$m_3 = 1$ kg

$F = 14$ N

$a = \frac{F}{m_1 + m_2 + m_3}$

$a = \frac{14}{4 + 2 + 1}$

$a = \frac{14}{7}$

$a = 2$ m/s^2

21. $T_2 = m_1 a$

$T_2 = 4 \times 2$

$T_2 = 8$ N

22. $T_1 = (m_1 + m_2)a = (4 + 2) \times 2 = 6 \times 2$

$T_1 = 12$ N

Level – III

23) B 24) D 25) A 26) B 27) A 28) C

29) a-r; b-q; c-p; d-p

23. $a = \left[\frac{m_2 - m_1}{m_1 + m_2}\right] \times g$

$a = \left[\frac{70 - 30}{30 + 70}\right] \times 10$

$a = \frac{40}{100} \times 10$

$a = 4\ m/s^2$

24. $T = \frac{2\ m_1\ m_2\ g}{m_1 + m_2}$

$T = \frac{2 \times 70 \times 30}{100} \times 10$

$T = 420\ N$

25. $S = \frac{1}{2} a t^2$

$1 = \frac{1}{2} \times 8 \times t^2$

$t^2 = 1/4$

$t = 1/\sqrt{4} sec$

$t = 0.5 sec$

WORKSHEET-6

Level – I

1) D 2) B 3) D 4) D 5) A

3. $W = m(g - a) = 50(9.8 - 1) = 440\ N$ MAINS & ADVANCED

Level – II

1) C 2) A 3) C 4) B 5) C 6) B 7) A

8) C 9) D 10) B 11) A 12) B 13) A 14) A

15) A 16) B

3. $m = 100\ kg$

$a = 10\ m/s^2$

$W^1 = m(g + a)$

$W^1 = 100(10 + 4)$

$W^1 = 100 \times 14 = 1400\ N$

$W^1 = m^1\ g$

$m^1 = \frac{w^1}{g} = \frac{1400}{10}$

$m^1 = 140\ kg$

5

$\mathrm{m}\ 1\ \mathrm{kg}m = 1 \times 10 = 10\ \mathrm{N}$

$\mathrm{m} = 1\ \mathrm{kg}$

$\mathrm{g} = 10\ \mathrm{m/s^2/s^2}$

$\mathrm{T} = \mathrm{w}^1 = \mathrm{m(g + aa)}$

$\mathrm{T} = 1(10 + 5)$

$\mathrm{T} = 15\ \mathrm{N}$

6. $\mathrm{T} = 25\ \mathrm{N}$

$\mathrm{a} = 2.5\ \mathrm{m/s^2}$

$\mathrm{g} = 10\ \mathrm{m/s^2}$

$\mathrm{T} = \mathrm{m(g + a)}$

$25 = \mathrm{m}(10 + 2.5)$

$\mathrm{m} = \frac{25}{12.5} = 2\ \mathrm{kg}$

7. $\mathrm{a} = \mathrm{g}/2$

$\mathrm{w_1} = \mathrm{m(g + a)}$

$= \mathrm{m(g + g_2)}$

$\mathrm{w_1} = \frac{3}{2}\mathrm{mg}$

$\mathrm{w_2} = \mathrm{m(g - a)}$

$\mathrm{w_2} = \frac{\mathrm{mg}}{2}$

$\frac{w_1}{w_2} = \frac{\frac{3}{2}mg}{\frac{1}{2}mg} = \frac{3}{1}$

8. $\mathrm{w}^1 = \frac{\mathrm{mg}}{4}$

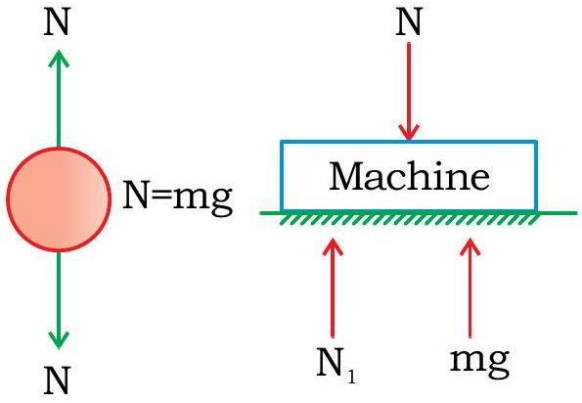
$\mathrm{w}^1 = \mathrm{m(g - a)}$

$\frac{\mathrm{mg}}{4} = \mathrm{m(g - a)}$

$g = 4g - 4a$

$4\mathrm{a} = 3\ \mathrm{g}\ a = \frac{3g}{4}$

12. For calculating the reading of weighing machine, we should draw F.B.D of man and machine seperately.



$\therefore N = mg = 60 \times 10 \Rightarrow 600\ N$

13. $R = m(g + a)$

$= 50 \times (10 + 2) = 600\ N$

$= 60kgwt$

Level – III

17) A 18) B 19) B 20) a-p; b-q; c-r; d-s

WORKSHEET-7

Level – I

1) B 2) C 3) D 4) B 5) A 6) D 7) C

8) A 9) D

Level – II

1) A 2) C 3) D 4) C 5) D 6) B 7) A

8) B 9) D 10) D 11) A 12) B

8. $f = 100g.wt = 100 \times 9.8\ kg.wt$

$R = mg, m = 1\ kg, g = 9.8m/s^2$

$f = \mu R; \mu = \frac{f}{R} = \frac{100 \times 9.8}{1000 \times 9.8} = \frac{1}{10} = 0.1$

11. m = 5 kg

u = 10 m/sv = 0 m/s

s = 10 m

from laws of motion $v^2 - u^2 = 2as$

$0^2 - 100 = 2as$

$a = -\frac{100}{20} = 5\ m/sec$

we know f = ma

$= 5 \times 5$

$f = 25N$

12. $f = 20\ \text{kg} \cdot \text{wt}; m = 50\ \text{kg};$

$$\mu = \frac{F}{R} = \frac{20 \times 9.8}{50 \times 9.8} = \frac{2}{5} = 0.4$$

$(1\ \text{kg} \cdot \text{wt} = 9.8\ \text{N})$.

Level – III

13) A	14) a-q,r; b-s; c-p;d-t	15) A	16) A	17) D	18) D
19) B	20) A,B,C	21) 4			

16. $m = 100\text{gm} = \frac{100}{1000} = \frac{1}{10}\ \text{kg}$

$f = 0.8\text{Ng} = 10\ \text{m/sec}^2$

$$\mu = \frac{f}{R} = \frac{0.8}{\frac{1}{10} \times 10} = 0.8$$

17. $R = 10\ \text{N}$

$w = f = \mu R$

$= 0.2 \times 10 = 2\ \text{N}$

20. Conceptual

21. $M = 100\text{gm} = \frac{1}{10}\ \text{kg}$

$f = 0.4\text{Ng} = 10\ \text{m/sec}^2$

$f = \mu R$

$$\mu = \frac{f}{R} = \frac{0.4}{\frac{1}{10} \times 10} = 0.4 = 4 \times \frac{1}{10}$$

$$= 4 \times 10^{-1}$$

WORKSHEET-8

Level – I

1) A	2) A	3) D	4) A	5) A

Level – II

1) C	2) D	3) B	4) B	5) C	6) A	7) B
8) C	9) A	10) A	11) A	12) A	13) A	14) A
15) B						

6. $F_B = F_g = 5.0\ \text{N}$

7. $W_{net} = F_g - V\rho g$

$$= 54\ \text{N} - (2.3 \times 10^{-3}\ \text{m}^3) \times (1.00 \times 10^3\ \text{kg/m}^3)(9.80\ \text{m/s}^2)$$

$$= 54\ \text{N} - 23\ \text{N} = 31\ \text{N}$$

11. $F_g = F_{buoyant} = \rho_{water}\, v_g$

$$v = \frac{F_g}{\rho_{water}\, g}$$

$$= \frac{600\ \text{N}}{(1000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)} = 0.06\ \text{m}^3$$

This volume does not include that portion of her head that is above the water.

12. The buoyant force will be twice as great as in water, or $2.00 \times 10^3\ \text{kg/m}^3$

$$W_{\text{net}} = 54\ \text{N} - (2.3 \times 10^{-3}\ \text{m}^3) \times (2.00 \times 10^3\ \text{kg/m}^3)(9.80\ \text{m/s}^2) = 8.9\ \text{N}$$

13. $F_T + F_{\text{buoyant}} = F_g$ where F_g is the air weight of the camera.

$$F_T = F_g - F_{\text{buoyant}} = F_g - \rho_{\text{water}}\, V_g$$

$$= 1250 - (1000\ \text{kg/m}^3)(0.08\ \text{m}^3)(9.80\ \text{m/s}^2)$$

$$= 4.4 \times 10^2\ \text{N}$$

14. $F_B = V\rho g = (0.0010\ \text{m}^3)(1000\ \text{kg/m}^3) \times (9.80\ \text{m/s}^2)$

$= 9.8\ \text{N}$

The buoyant force depends only on the volume of the water displaced.

15. $F_{\text{net}} = F_g - F_{\text{buoyant}} = V\rho_{\text{He}} g - V\rho_{\text{air}}\, g = (1.00\ \text{m}^3)(0.177\ \text{kg/m}^3)(9.80\ \text{m/s}^2)$

$-(1.00\ \text{m}^3)(1.20\ \text{kg/m}^3)(9.80\ \text{m/s}^2)$

$= -10.0\ \text{N}$ (Net buoyant force is 10.0 N upward.)

$F_{\text{max}} = 10.0\ \text{N}$

Level – III

16) B,C,D 17) C 18) D 19) A 20) C 21) A

22) A 23) A

22. $F_{\text{buoyant}} = V\rho_{\text{water}}\, g$

$$V = \frac{F_{\text{buoyant}}}{\rho_{\text{water}}\, g}$$

$$= \frac{920\ \text{N} - 750\ \text{N}}{(1.00 \times 10^3\ \text{kg/m}^3)(9.80\ \text{m/s}^2)}$$

$$= 1.7 \times 10^{-2}\ \text{m}^3$$

23. $\rho_{\text{object}} = \frac{m}{V} = \frac{F_g}{Vg}$

$$= \frac{920\ \text{N}}{(1.73 \times 10^{-2}\ \text{m}^3)(9.80\ \text{m/s}^2)}$$

$$= 5.4 \times 10^3\ \text{kg/m}^3$$